

project!!!!

```
# RUN THESE ALL ONCE!!! un-comment the code blocks as you go
#install.packages(c("usethis", "gitcreds"))

# then run this:
#usethis::create_github_token()

# which opens GitHub in your browser, generate a token
# COPY it before closing the page

# run this and paste token into terminal when prompted
#gitcreds::gitcreds_set()

# done! :) make sure to re-comment the code blocks
```

Question 1 : Exploratory Analysis and Model Specification

The Demographic data set was first explored using descriptive statistics and correlation analyses in order to understand the relationships between crime rates and demographic variables across four regions. The data set was split into four regions, region 1 containing 103 observations, region 2 containing 108 observations, region 3 containing 152 observations and region 4 containing 77 observations. Summary statistics were used to explore and evaluate the ranges, distributions, and variability of the demographic and socioeconomic variables.

The correlation matrices and plots were used to examine the relationships between the variables. If a variable contained a stronger correlation to the response variable, they were selected as potential predictors for the regional regression models. Separate analyses was done for each region to determine whether different predictors were associated with crime rates. This analysis suggested that different demographic variables were more strongly associated with crime depending on the region. In this project, we selected total population, the number of physicians, personal income, and population of ages 18 to 34.

```
Demographic <- read.table(here("data", "Demographic.txt")) # more robust read-in

# change column names to match correct variables
colnames(Demographic) <- c('ID', 'County', 'state', 'Land_area', 'total_population',
                           'Population_18to34', 'Population_65', 'Physicians', 'beds',
                           'Y_i', 'Graduate_highschool', 'Graduate_Bachelor', 'below_poverty',
                           'unemployment', 'capita_income', 'personal_income', 'Geographic_region')

Demographic2 <- Demographic

# Variable names and dimensions of the table
head(Demographic2)
```

```
##   ID      County state Land_area total_population Population_18to34
## 1   1 Los_Angeles   CA      4060      8863164           32.1
```

```
## 2 2 Cook IL 946 5105067 29.2
## 3 3 Harris TX 1729 2818199 31.3
## 4 4 San_Diego CA 4205 2498016 33.5
## 5 5 Orange CA 790 2410556 32.6
## 6 6 Kings NY 71 2300664 28.3
## Population_65 Physicians beds Y_i Graduate_highschool Graduate_Bachelor
## 1 9.7 23677 27700 688936 70.0 22.3
## 2 12.4 15153 21550 436936 73.4 22.8
## 3 7.1 7553 12449 253526 74.9 25.4
## 4 10.9 5905 6179 173821 81.9 25.3
## 5 9.2 6062 6369 144524 81.2 27.8
## 6 12.4 4861 8942 680966 63.7 16.6
## below_poverty unemployment capita_income personal_income Geographic_region
## 1 11.6 8.0 20786 184230 4
## 2 11.1 7.2 21729 110928 2
## 3 12.5 5.7 19517 55003 3
## 4 8.1 6.1 19588 48931 4
## 5 5.2 4.8 24400 58818 4
## 6 19.5 9.5 16803 38658 1
```

```
names(Demographic2)
```

```
## [1] "ID" "County" "state"
## [4] "Land_area" "total_population" "Population_18to34"
## [7] "Population_65" "Physicians" "beds"
## [10] "Y_i" "Graduate_highschool" "Graduate_Bachelor"
## [13] "below_poverty" "unemployment" "capita_income"
## [16] "personal_income" "Geographic_region"
```

```
dim(Demographic2)
```

```
## [1] 440 17
```

```
summary(Demographic2)
```

```
## ID County state Land_area
## Min. : 1.0 Length:440 Length:440 Min. : 15.0
## 1st Qu.:110.8 Class :character Class :character 1st Qu.: 451.2
## Median :220.5 Mode :character Mode :character Median : 656.5
## Mean :220.5 Mean : 1041.4
## 3rd Qu.:330.2 3rd Qu.: 946.8
## Max. :440.0 Max. :20062.0
## total_population Population_18to34 Population_65 Physicians
## Min. : 100043 Min. :16.40 Min. : 3.000 Min. : 39.0
## 1st Qu.: 139027 1st Qu.:26.20 1st Qu.: 9.875 1st Qu.: 182.8
## Median : 217280 Median :28.10 Median :11.750 Median : 401.0
## Mean : 393011 Mean :28.57 Mean :12.170 Mean : 988.0
## 3rd Qu.: 436064 3rd Qu.:30.02 3rd Qu.:13.625 3rd Qu.: 1036.0
## Max. :8863164 Max. :49.70 Max. :33.800 Max. :23677.0
## beds Y_i Graduate_highschool Graduate_Bachelor
## Min. : 92.0 Min. : 563 Min. :46.60 Min. : 8.10
## 1st Qu.: 390.8 1st Qu.: 6220 1st Qu.:73.88 1st Qu.:15.28
```

```
## Median : 755.0 Median : 11820 Median :77.70 Median :19.70
## Mean : 1458.6 Mean : 27112 Mean :77.56 Mean :21.08
## 3rd Qu.: 1575.8 3rd Qu.: 26280 3rd Qu.:82.40 3rd Qu.:25.32
## Max. :27700.0 Max. :688936 Max. :92.90 Max. :52.30
## below_poverty unemployment capita_income personal_income
## Min. : 1.400 Min. : 2.200 Min. : 8899 Min. : 1141
## 1st Qu.: 5.300 1st Qu.: 5.100 1st Qu.:16118 1st Qu.: 2311
## Median : 7.900 Median : 6.200 Median :17759 Median : 3857
## Mean : 8.721 Mean : 6.597 Mean :18561 Mean : 7869
## 3rd Qu.:10.900 3rd Qu.: 7.500 3rd Qu.:20270 3rd Qu.: 8654
## Max. :36.300 Max. :21.300 Max. :37541 Max. :184230
## Geographic_region
## Min. :1.000
## 1st Qu.:2.000
## Median :3.000
## Mean :2.461
## 3rd Qu.:3.000
## Max. :4.000
```

```
Demographic2$Y_i <- log(Demographic$Y_i) # log transform Y
```

```
colnames(Demographic2)[10] <- "LogY_i"
```

```
region1 <- Demographic2[Demographic2$Geographic_region == 1,]
region2 <- Demographic2[Demographic2$Geographic_region == 2,]
region3 <- Demographic2[Demographic2$Geographic_region == 3,]
region4 <- Demographic2[Demographic2$Geographic_region == 4,]
```

```
n1<-nrow(region1)
n2<-nrow(region2)
n3<-nrow(region3)
n4<-nrow(region4)
```

```
# general corr
```

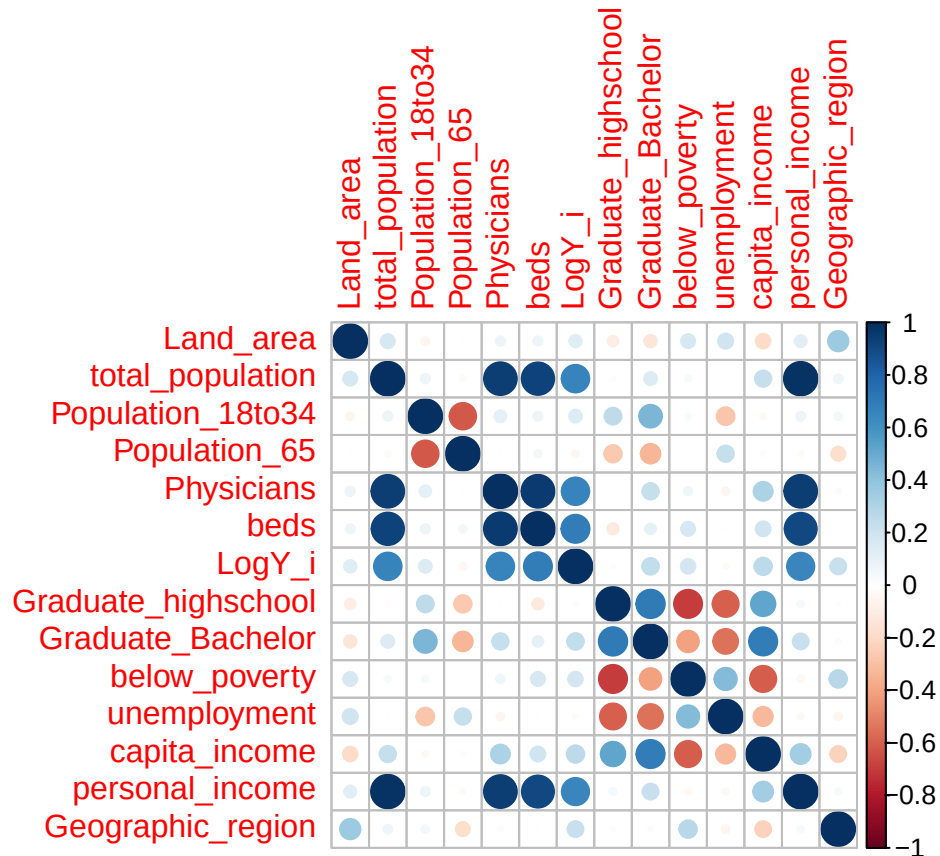
```
cor(Demographic2[,4:17])
```

```
## Land_area total_population Population_18to34
## Land_area 1.000000000 0.173083353 -0.05487812
## total_population 0.173083353 1.000000000 0.07837212
## Population_18to34 -0.054878125 0.078372117 1.00000000
## Population_65 0.005770871 -0.029037393 -0.61630964
## Physicians 0.078074657 0.940248591 0.11969924
## beds 0.073047270 0.923738360 0.07453191
## LogY_i 0.135368016 0.667132590 0.14545331
## Graduate_highschool -0.098598111 -0.017426900 0.25058429
## Graduate_Bachelor -0.137237736 0.146813850 0.45609703
## below_poverty 0.171343348 0.038019509 0.03397551
## unemployment 0.199209277 0.005351703 -0.27852706
## capita_income -0.187715132 0.235610188 -0.03164843
## personal_income 0.127074261 0.986747626 0.07116151
## Geographic_region 0.362868243 0.069437072 0.05241407
## Population_65 Physicians beds LogY_i
## Land_area 0.005770871 0.078074657 0.073047270 0.13536802
```

## total_population	-0.029037393	0.940248591	0.923738360	0.66713259
## Population_18to34	-0.616309639	0.119699240	0.074531907	0.14545331
## Population_65	1.000000000	-0.003128630	0.053278417	-0.04383737
## Physicians	-0.003128630	1.000000000	0.950464395	0.66518078
## beds	0.053278417	0.950464395	1.000000000	0.69141343
## LogY_i	-0.043837372	0.665180779	0.691413431	1.00000000
## Graduate_highschool	-0.268251758	-0.004248085	-0.111916382	-0.01866928
## Graduate_Bachelor	-0.339228765	0.236765466	0.100426534	0.24230414
## below_poverty	0.006578474	0.064136254	0.172793840	0.18596498
## unemployment	0.236309411	-0.050516116	0.007523992	-0.02154784
## capita_income	0.018590706	0.316134625	0.194808180	0.26899447
## personal_income	-0.022733151	0.948110571	0.902061545	0.65106347
## Geographic_region	-0.173291567	0.024646071	-0.003106920	0.22355922
##	Graduate_highschool	Graduate_Bachelor	below_poverty	
## Land_area	-0.098598111	-0.13723774	0.171343348	
## total_population	-0.017426900	0.14681385	0.038019509	
## Population_18to34	0.250584290	0.45609703	0.033975512	
## Population_65	-0.268251758	-0.33922877	0.006578474	
## Physicians	-0.004248085	0.23676547	0.064136254	
## beds	-0.111916382	0.10042653	0.172793840	
## LogY_i	-0.018669276	0.24230414	0.185964984	
## Graduate_highschool	1.000000000	0.70778672	-0.691750483	
## Graduate_Bachelor	0.707786723	1.00000000	-0.408423848	
## below_poverty	-0.691750483	-0.40842385	1.000000000	
## unemployment	-0.593595788	-0.54090691	0.436947236	
## capita_income	0.522996133	0.69536186	-0.601725039	
## personal_income	0.043355729	0.22223013	-0.038739339	
## Geographic_region	-0.010055061	0.02029897	0.270984846	
##	unemployment	capita_income	personal_income	
## Land_area	0.199209277	-0.18771513	0.12707426	
## total_population	0.005351703	0.23561019	0.98674763	
## Population_18to34	-0.278527058	-0.03164843	0.07116151	
## Population_65	0.236309411	0.01859071	-0.02273315	
## Physicians	-0.050516116	0.31613462	0.94811057	
## beds	0.007523992	0.19480818	0.90206155	
## LogY_i	-0.021547839	0.26899447	0.65106347	
## Graduate_highschool	-0.593595788	0.52299613	0.04335573	
## Graduate_Bachelor	-0.540906913	0.69536186	0.22223013	
## below_poverty	0.436947236	-0.60172504	-0.03873934	
## unemployment	1.000000000	-0.32214439	-0.03387633	
## capita_income	-0.322144395	1.00000000	0.34768161	
## personal_income	-0.033876330	0.34768161	1.00000000	
## Geographic_region	-0.054378572	-0.22249375	0.03768546	
##	Geographic_region			
## Land_area	0.36286824			
## total_population	0.06943707			
## Population_18to34	0.05241407			
## Population_65	-0.17329157			
## Physicians	0.02464607			
## beds	-0.00310692			
## LogY_i	0.22355922			
## Graduate_highschool	-0.01005506			
## Graduate_Bachelor	0.02029897			
## below_poverty	0.27098485			

```
## unemployment          -0.05437857
## capita_income          -0.22249375
## personal_income        0.03768546
## Geographic_region      1.00000000
```

```
corrplot(cor(Demographic2[,4:17]))
```



```
# per region corr
regions <- list(region1, region2, region3, region4)
top_list <- list()

for (i in seq_along(regions)) {

  cat("FOR REGION: ", i, "\n")

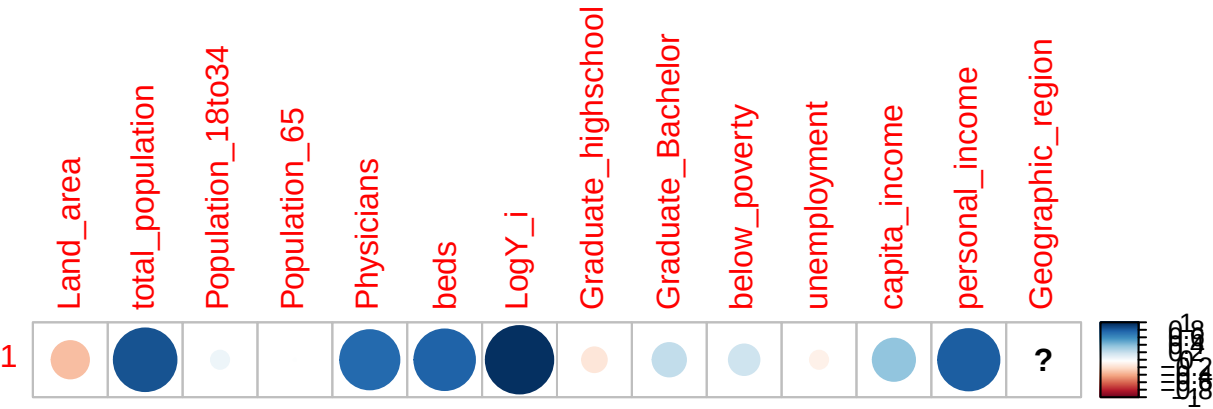
  region <- regions[[i]]
  cors <- sapply(region[, 4:17], function(x) cor(x, region$LogY_i))
  ranked <- sort(abs(cors), decreasing = TRUE)

  top_list[[i]] <- head(ranked, 5)

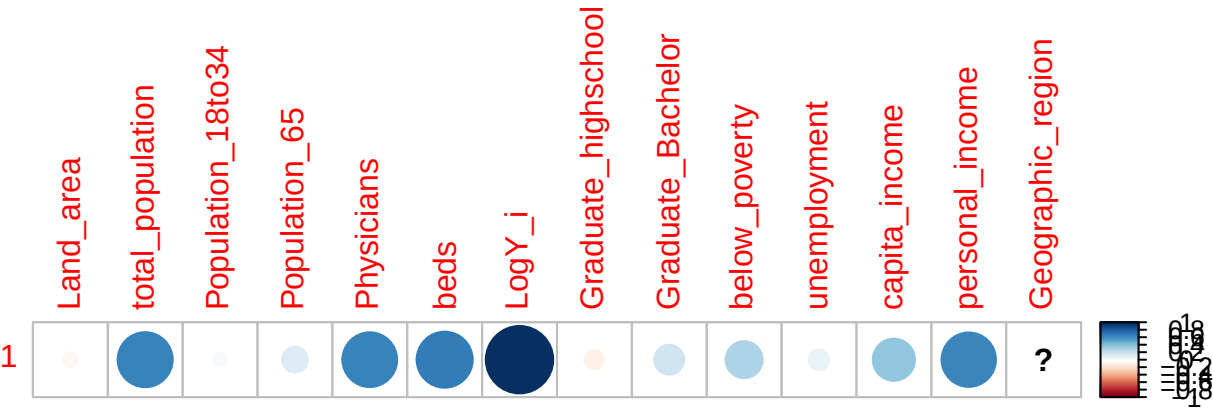
  corrplot(cor(region$LogY_i, region[,4:17]))

  print(top_list[[i]])
}
```

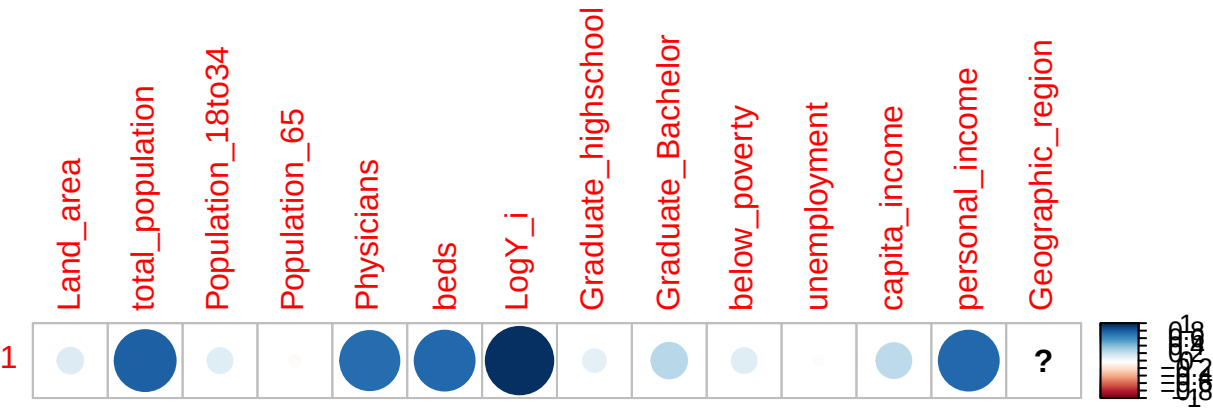
FOR REGION: 1



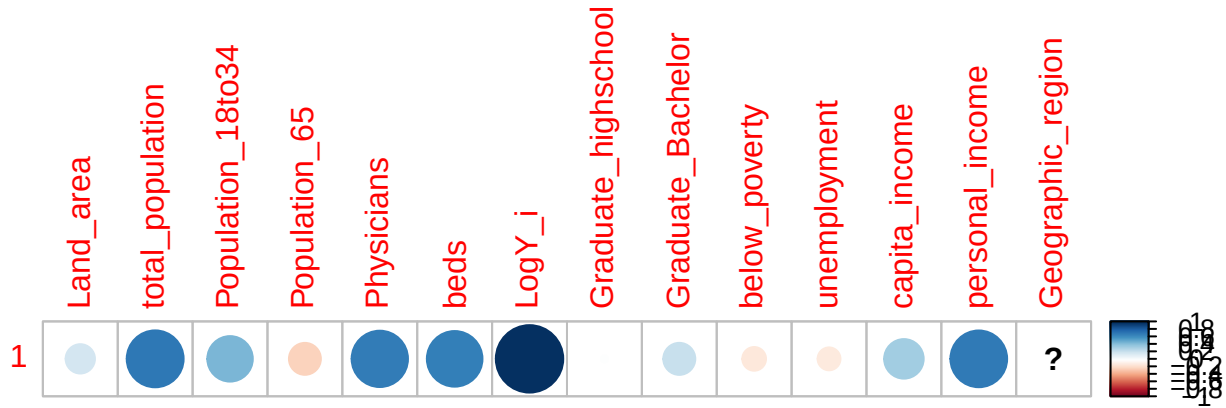
```
##          LogY_i total_population personal_income      beds
##          1.0000000      0.8625570      0.8255877      0.8068340
##          Physicians
##          0.7702250
## FOR REGION: 2
```



```
##          LogY_i      beds total_population      Physicians
##          1.0000000      0.6919176      0.6679155      0.6608150
##          personal_income
##          0.6518654
## FOR REGION: 3
```



```
##          LogY_i total_population personal_income      beds
##          1.0000000      0.8155360      0.7879222      0.7878209
##          Physicians
##          0.7689487
## FOR REGION: 4
```

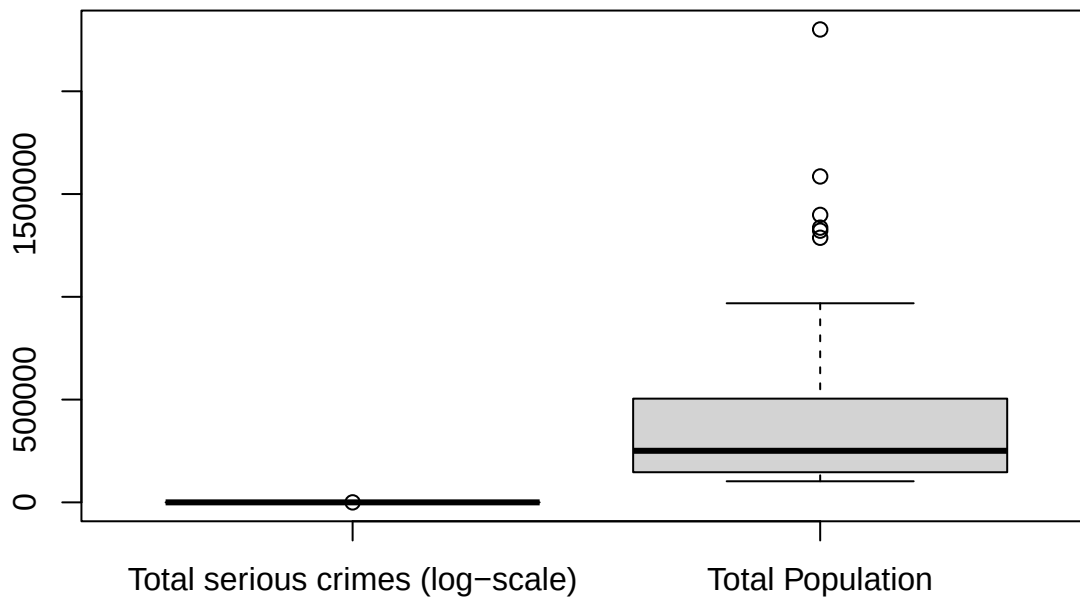


```
##          LogY_i total_population personal_income      Physicians
##          1.0000000      0.7120265      0.7078830      0.6977282
##          beds
##          0.6803344
```

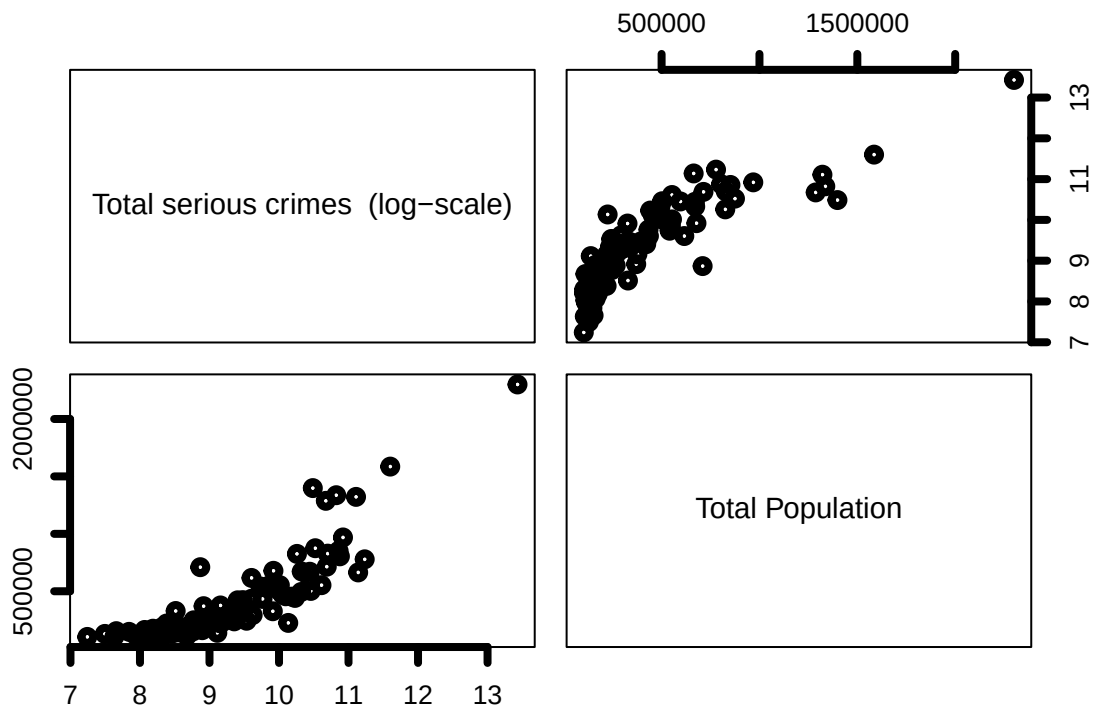
Region 1

```
# log(Y_i) = Beta_0 + Beta_1*total_population
```

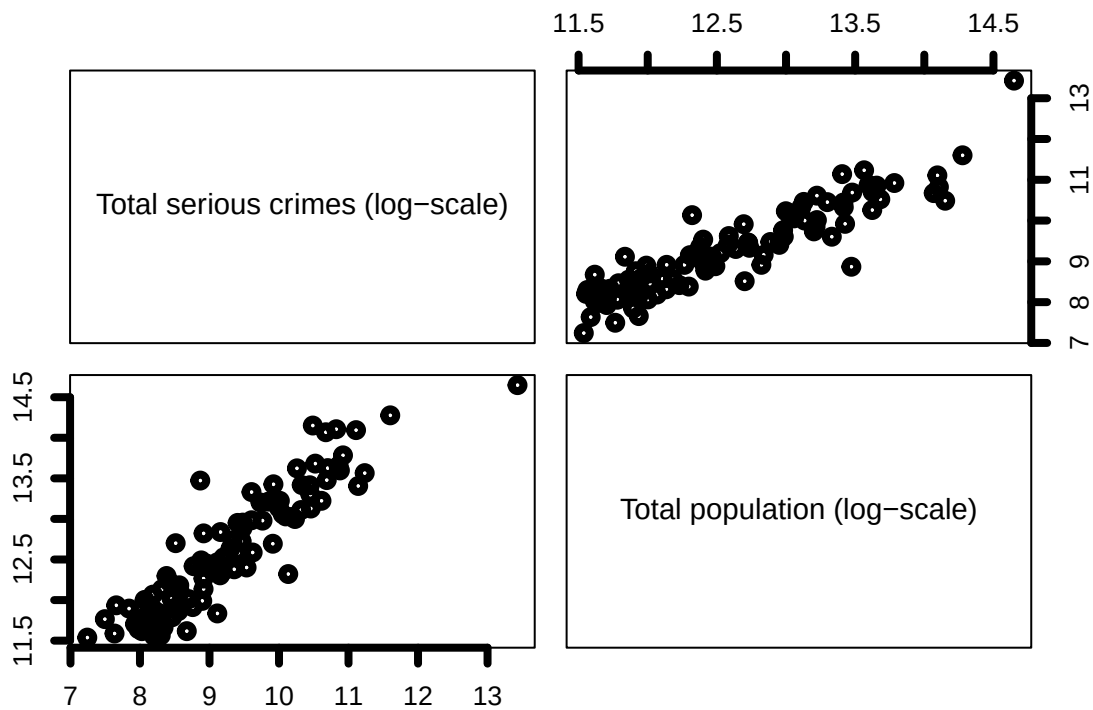
```
boxplot(cbind(region1$LogY_i, region1$total_population), names = c("Total serious crimes (log-scale)", "Total Population"))
```



```
pairs(cbind(region1$LogY_i, region1$total_population), labels=c("Total serious crimes (log-scale)", "To
```



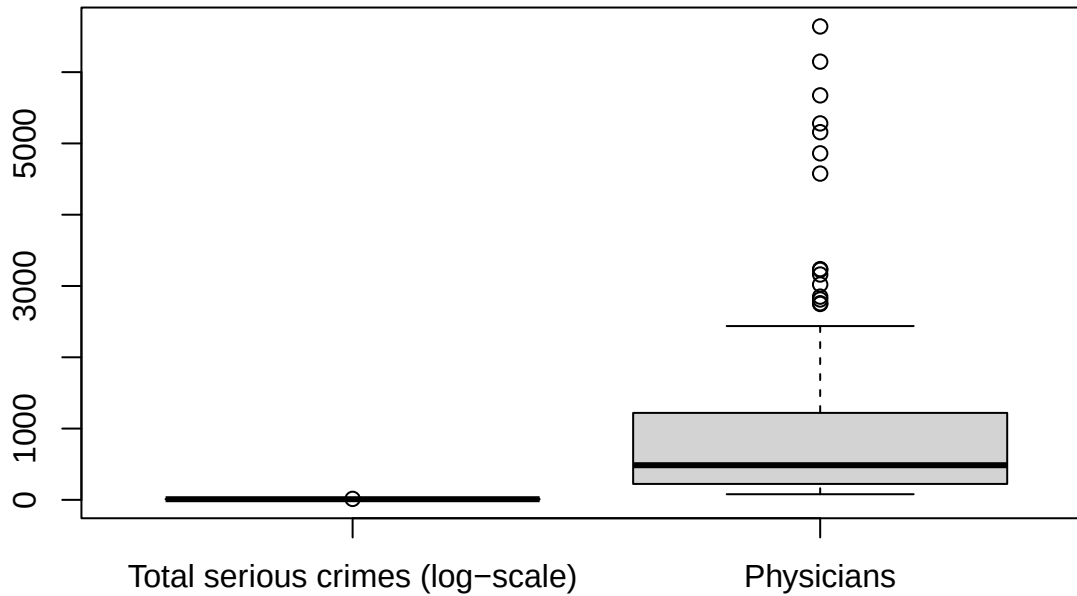
```
pairs(cbind(region1$LogY_i, log(region1$total_population)), labels = c("Total serious crimes (log-scale)", "To
```



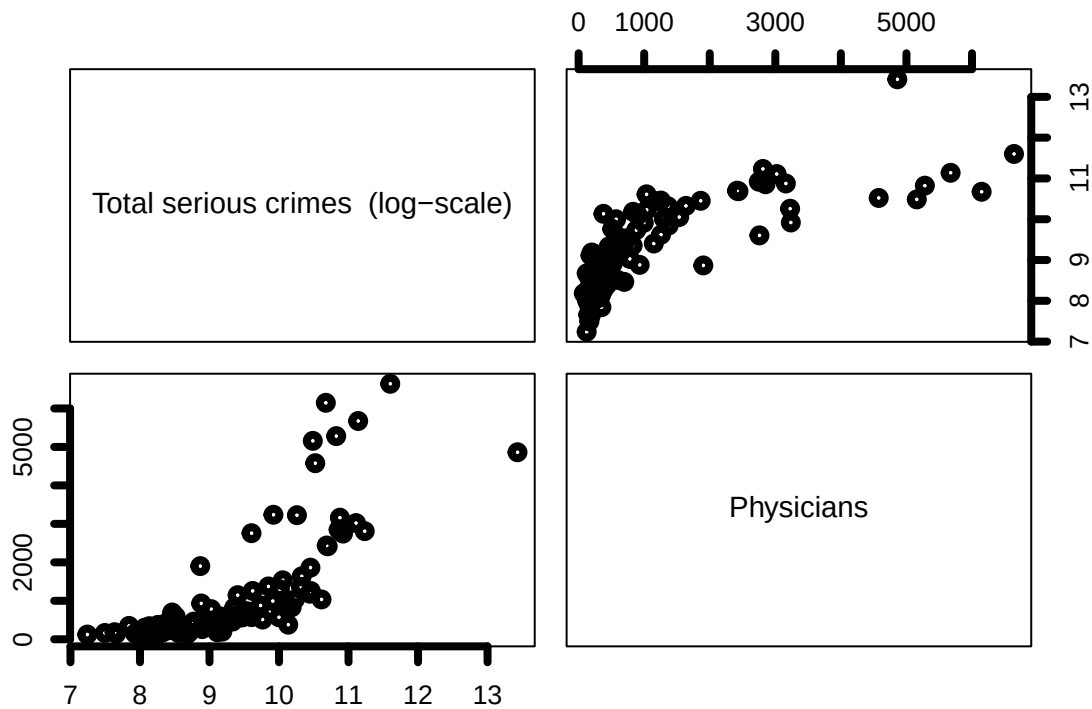
Region 2

```
#  $\log(Y_i) = \text{Beta}_0 + \text{Beta}_1 * \text{total\_population}$ 
```

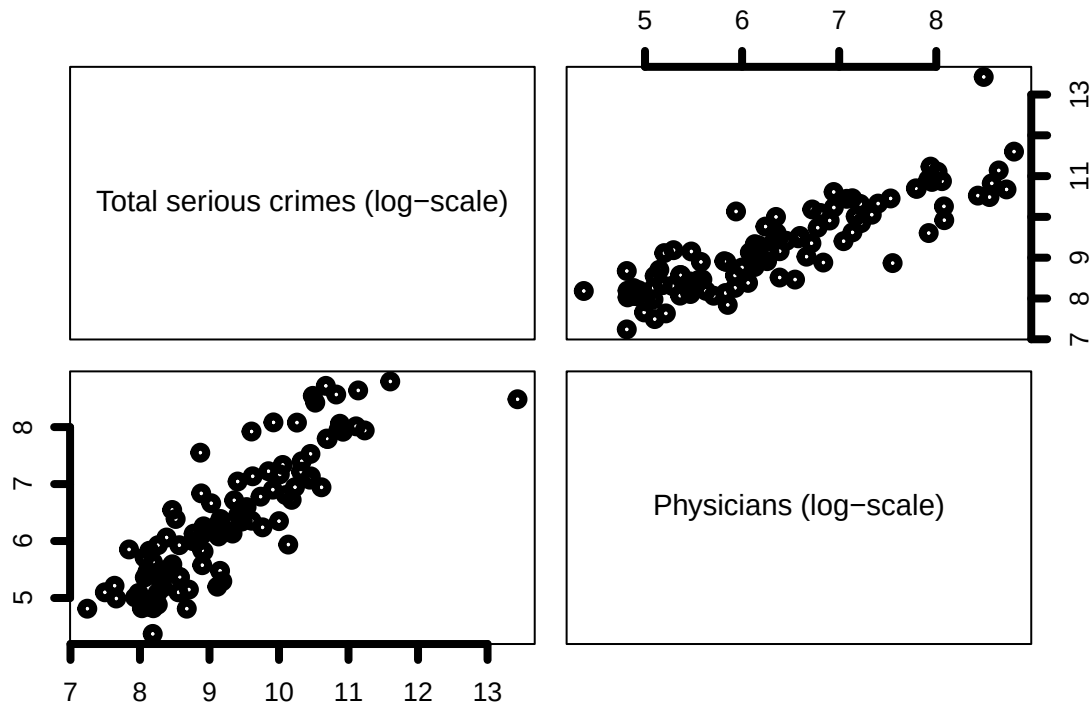
```
boxplot(cbind(region1$LogY_i, region1$Physicians), names = c("Total serious crimes (log-scale)", "Physicians"))
```



```
pairs(cbind(region1$LogY_i, region1$Physicians), labels=c("Total serious crimes (log-scale)", "Physicians"))
```



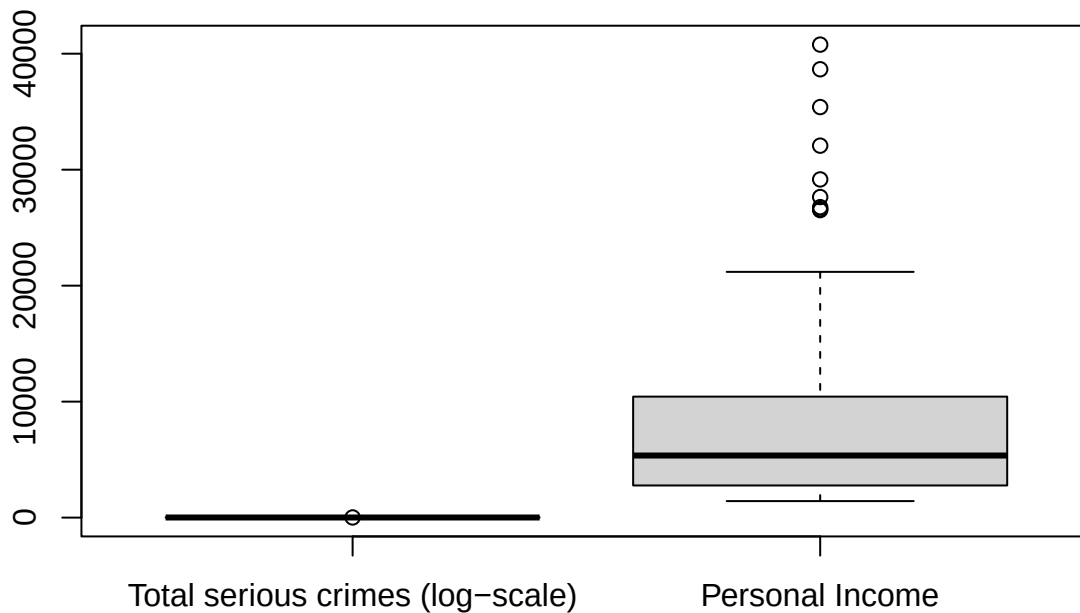
```
pairs(cbind(region1$LogY_i, log(region1$Physicians)), labels = c("Total serious crimes (log-scale)", "P
```



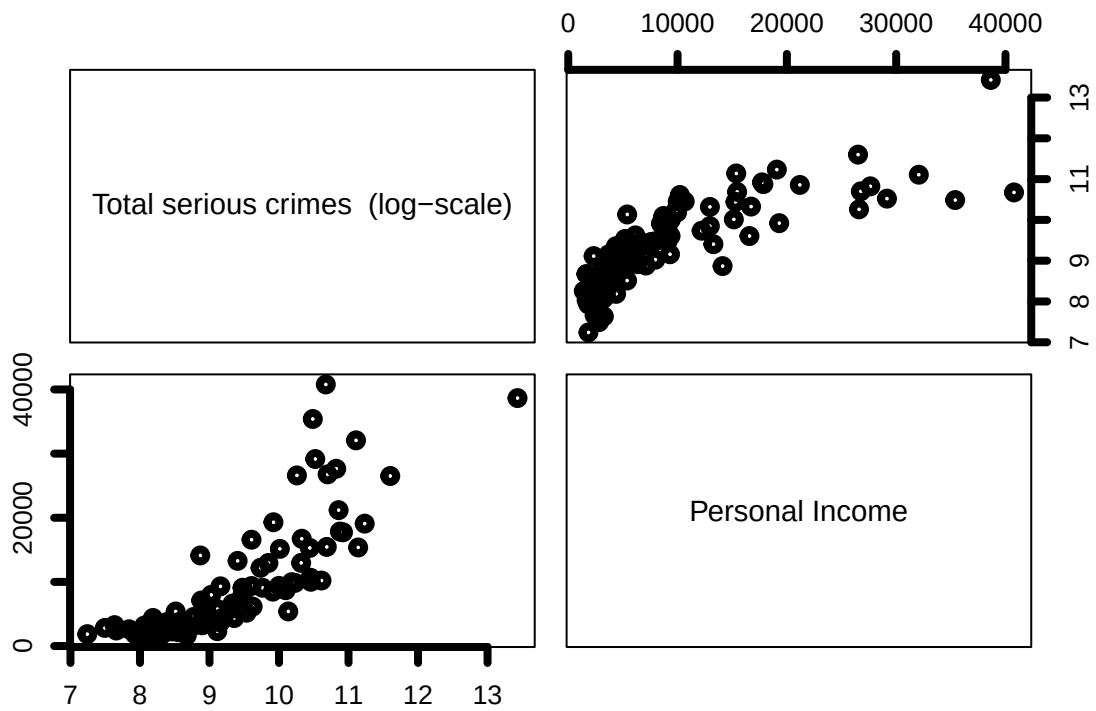
Region 3

```
# log(Y_i) = Beta_0 + Beta_1*total_population
```

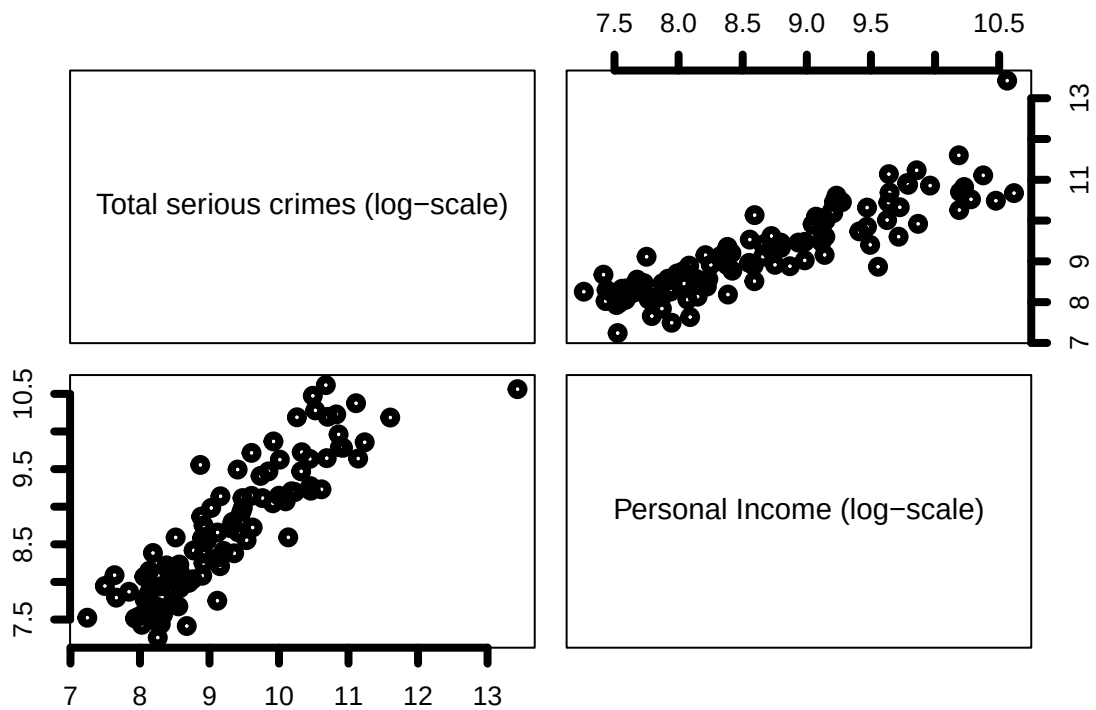
```
boxplot(cbind(region1$LogY_i, region1$personal_income), names = c("Total serious crimes (log-scale)", "P
```



```
pairs(cbind(region1$LogY_i, region1$personal_income), labels=c("Total serious crimes (log-scale)", "Per
```



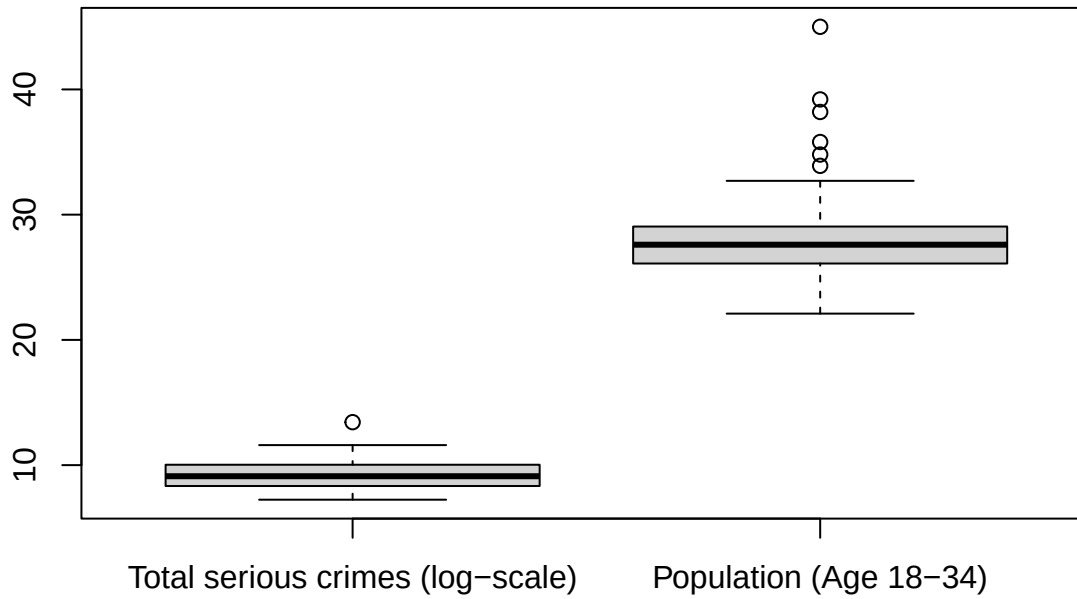
```
pairs(cbind(region1$LogY_i, log(region1$personal_income)), labels = c("Total serious crimes (log-scale)", "Per
```



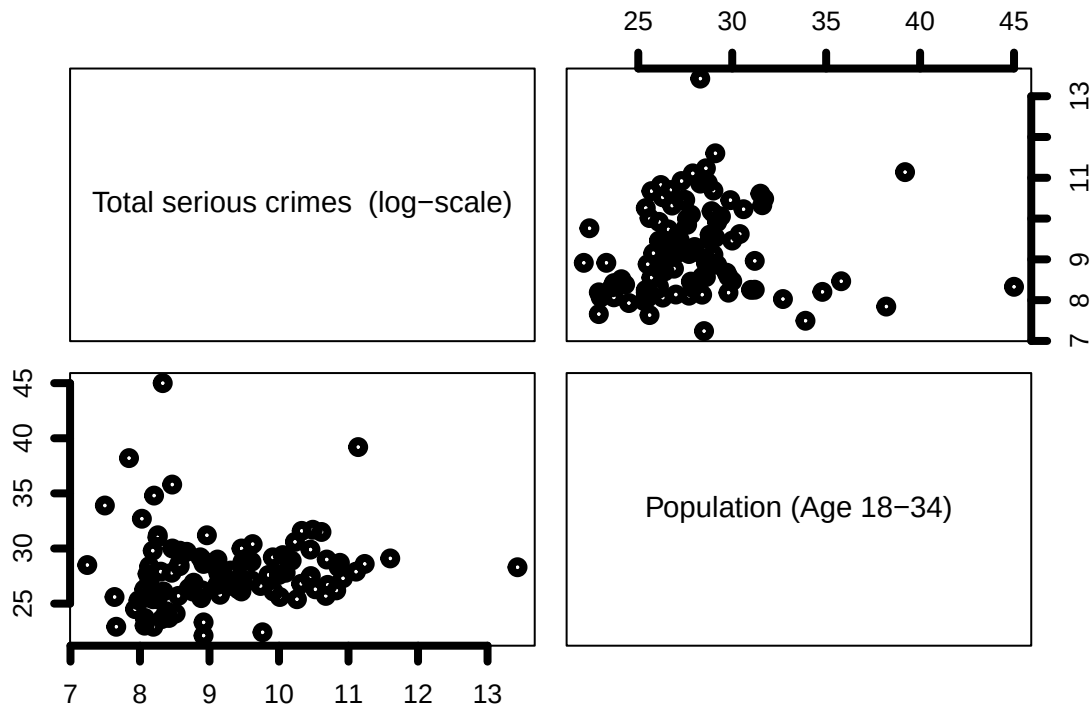
Region 4

```
# Region 1
# log(Y_i) = Beta_0 + Beta_1*total_population

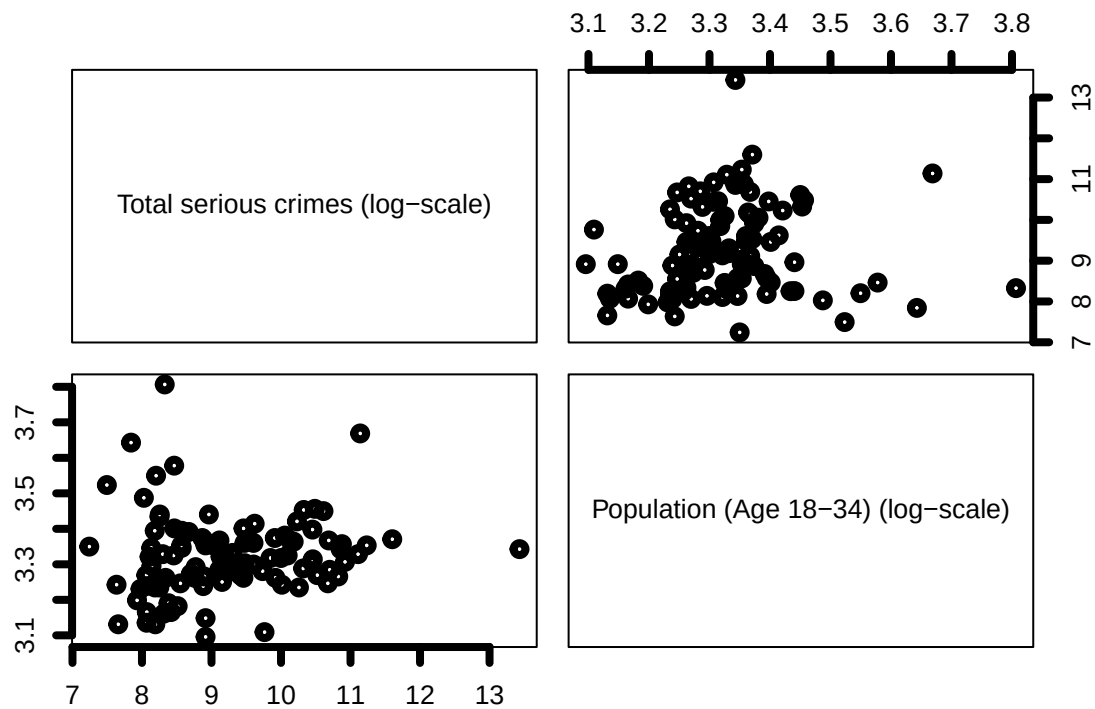
boxplot(cbind(region1$LogY_i, region1$Population_18to34), names = c("Total serious crimes (log-scale)",
```



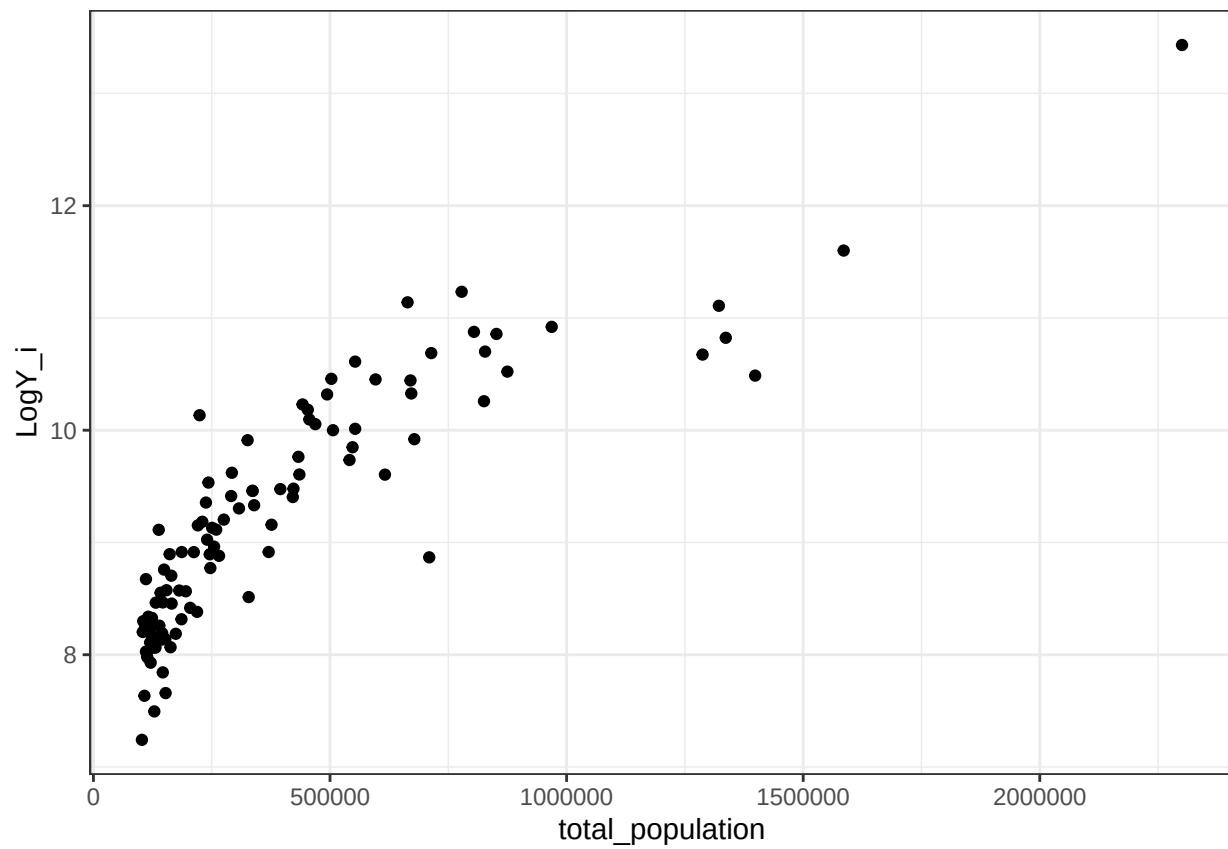
```
pairs(cbind(region1$LogY_i, region1$Population_18to34), labels=c("Total serious crimes (log-scale)", "P
```



```
pairs(cbind(region1$LogY_i, log(region1$Population_18to34)), labels = c("Total serious crimes (log-scale)", "Population (Age 18-34) (log-scale)"))
```



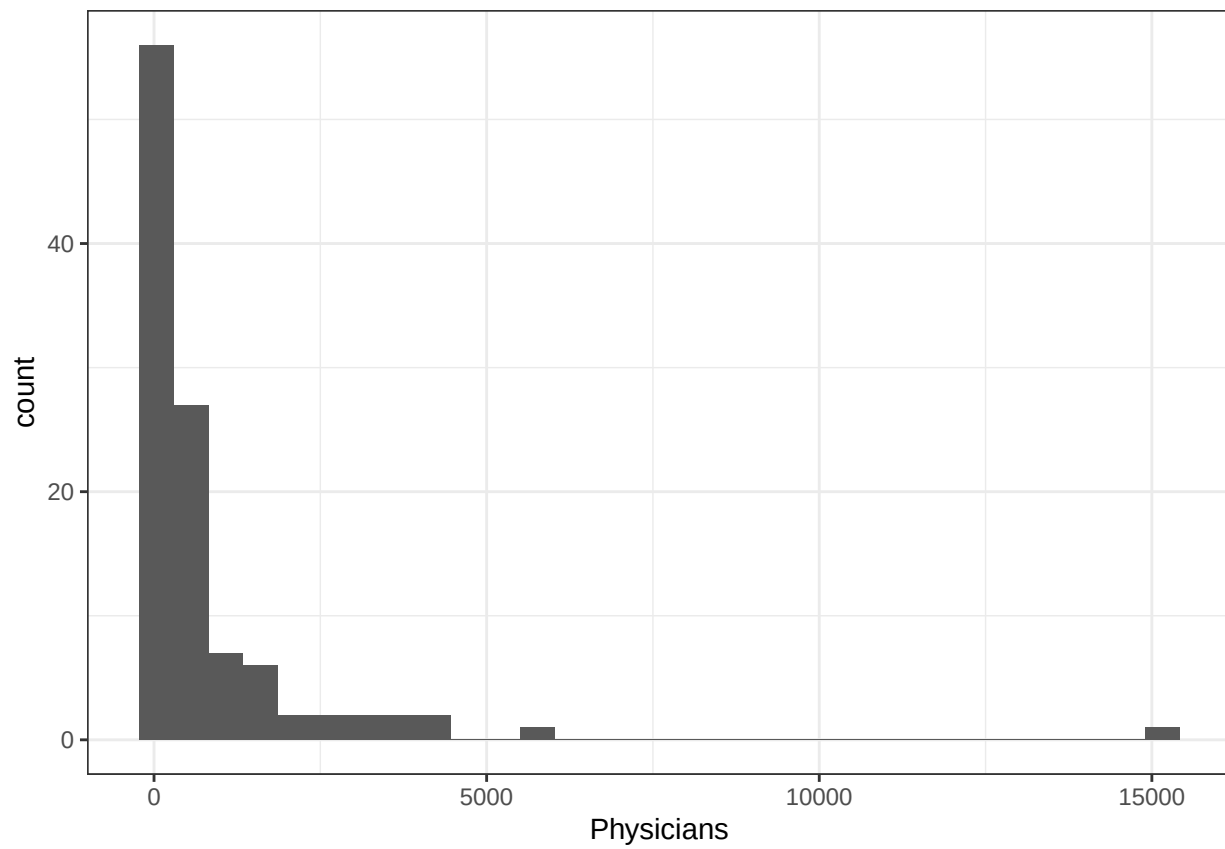
```
# Total Population
region1 |>
  ggplot(aes(x = total_population, y = LogY_i)) +
  geom_point() +
  theme_bw()
```



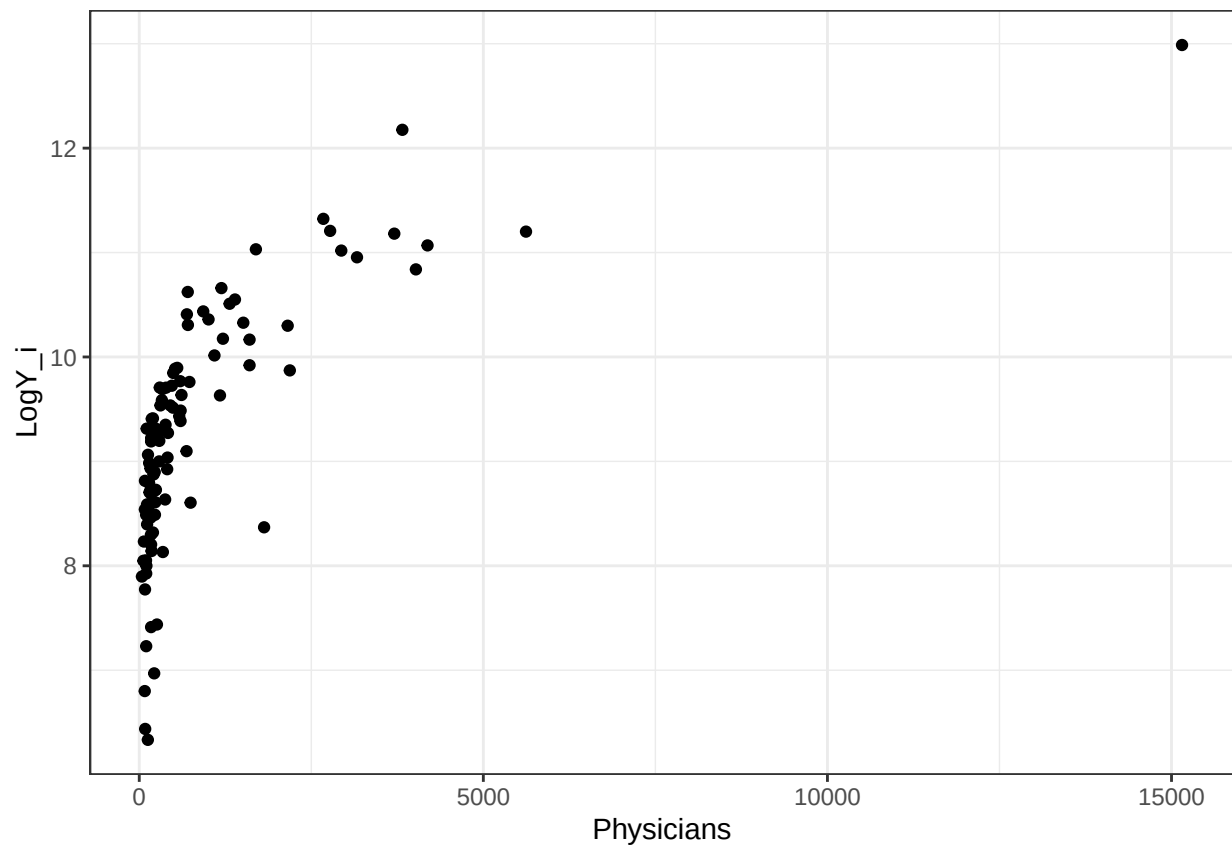
```
# Physicians
```

```
region2 |>  
  ggplot(aes(x = Physicians)) +  
  geom_histogram() +  
  theme_bw()
```

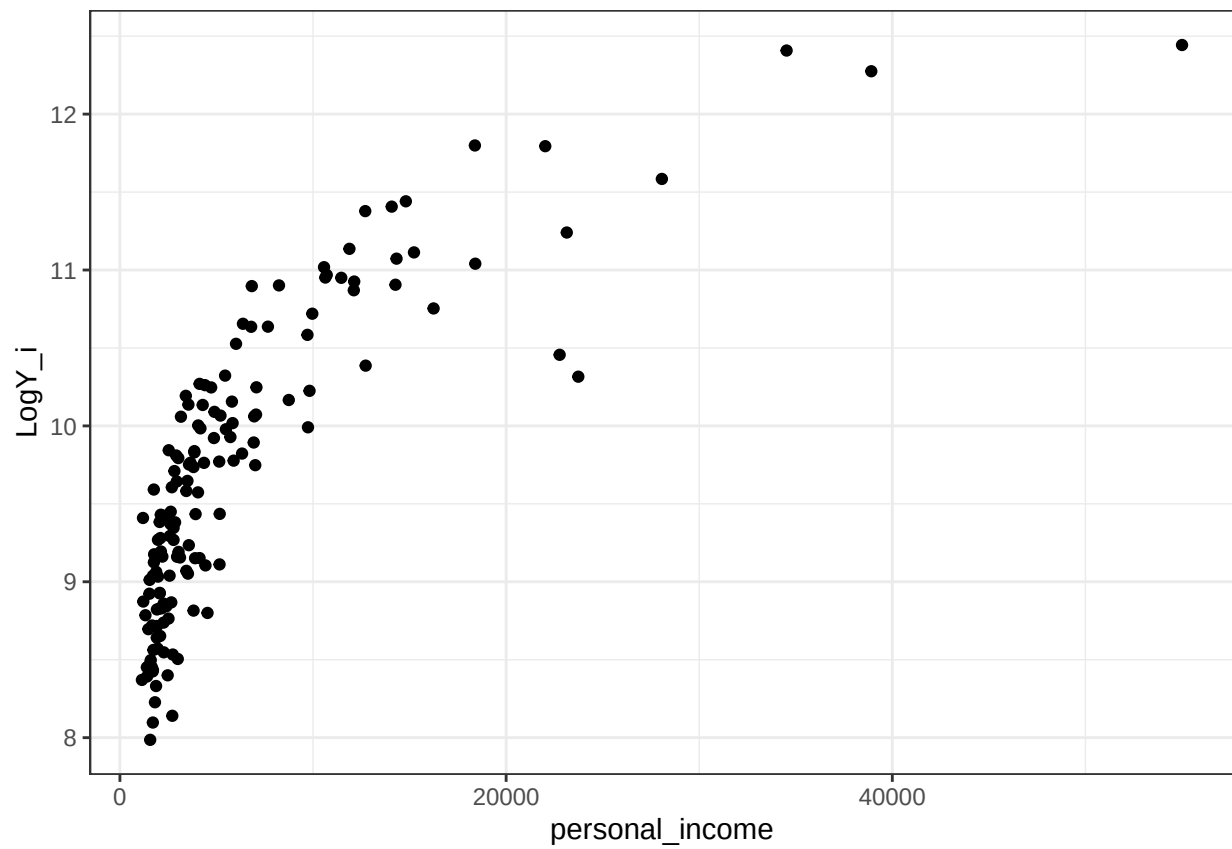
```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



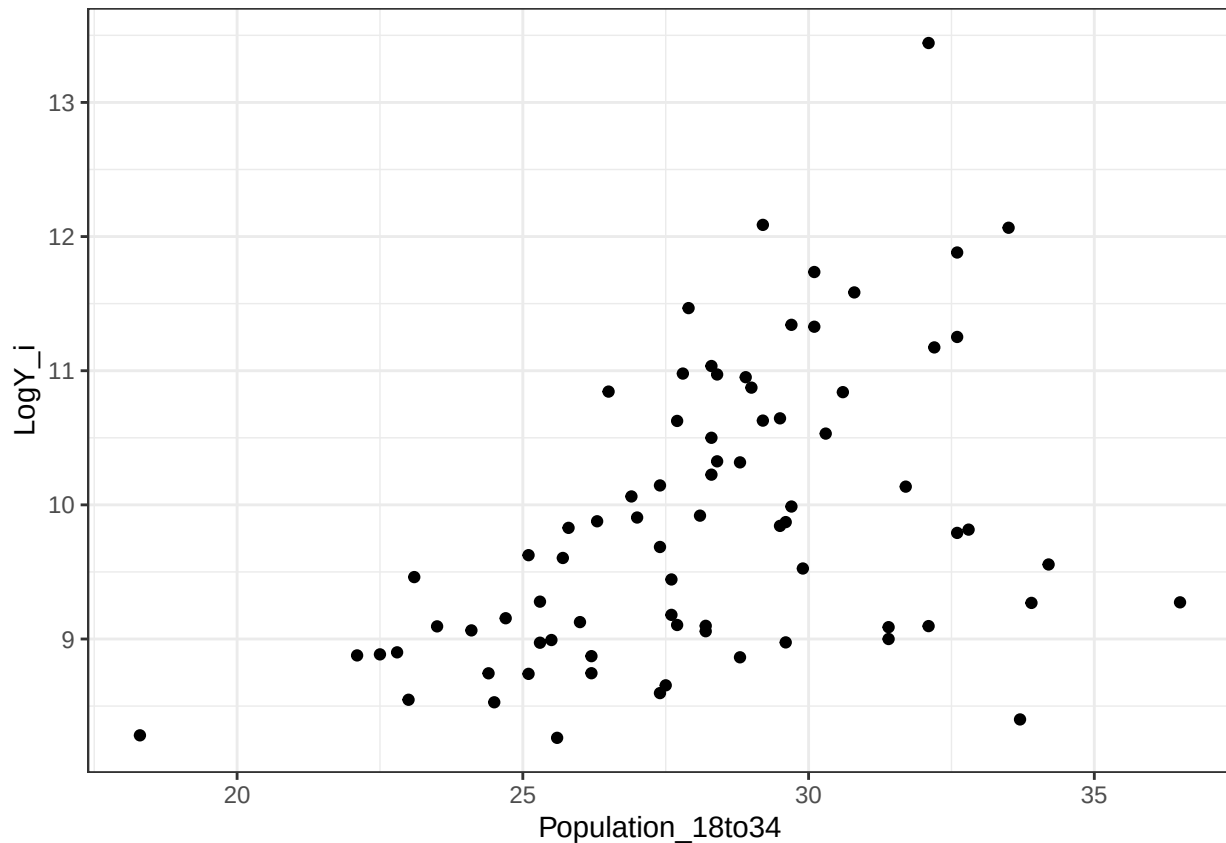
```
region2 |>  
  ggplot(aes(x = Physicians, y = LogY_i)) +  
  geom_point() +  
  theme_bw()
```



```
# Personal Income
region3 |>
  ggplot(aes(x = personal_income, y = LogY_i)) +
  geom_point() +
  theme_bw()
```

```
# Population 18-34  
region4 |>  
  ggplot(aes(x = Population_18to34, y = LogY_i)) +  
  geom_point() +  
  theme_bw()
```



CHOSEN PREDICTORS for SLR (no repeats):

Region 1: Total Population

Region 2: Physicians

Region 3: Personal Income

Region 4: Population 18-34

```
model_1 <- lm(LogY_i ~ total_population, data = region1)
model_2 <- lm(LogY_i ~ Physicians, data = region2)
model_3 <- lm(LogY_i ~ personal_income, data = region3)
model_4 <- lm(LogY_i ~ Population_18to34, data = region4)
```

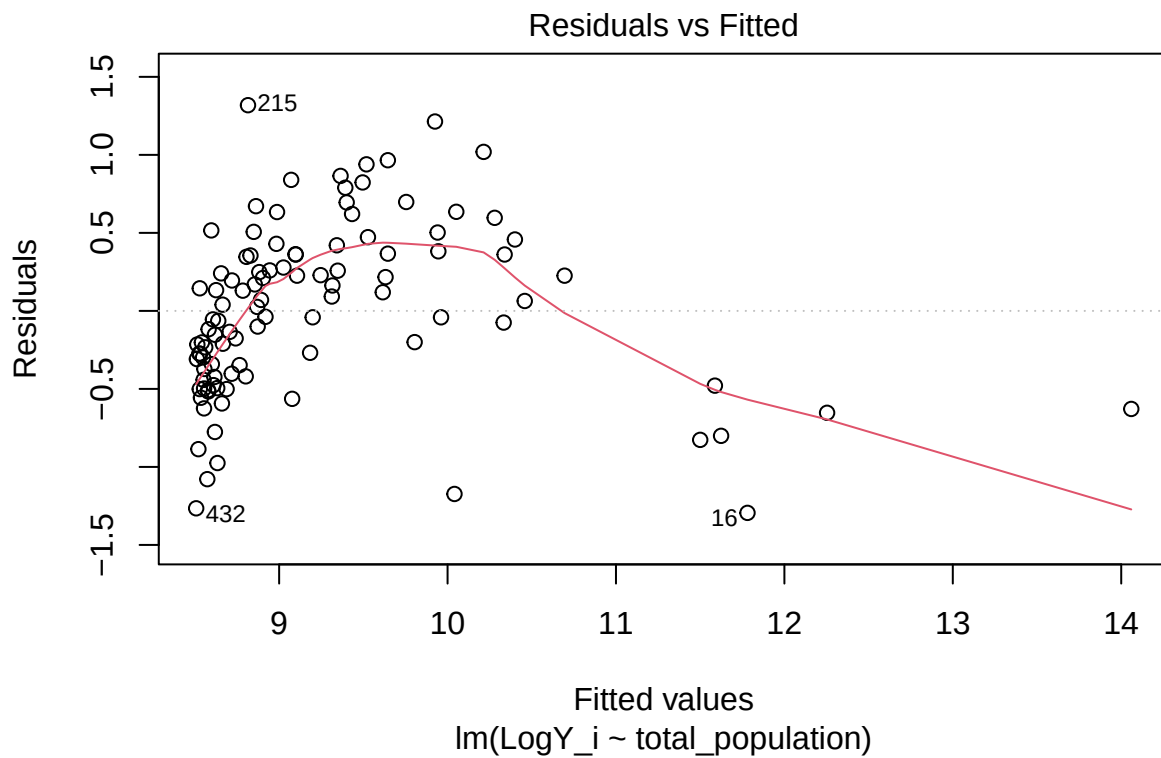
Question 2 : Model Evaluation

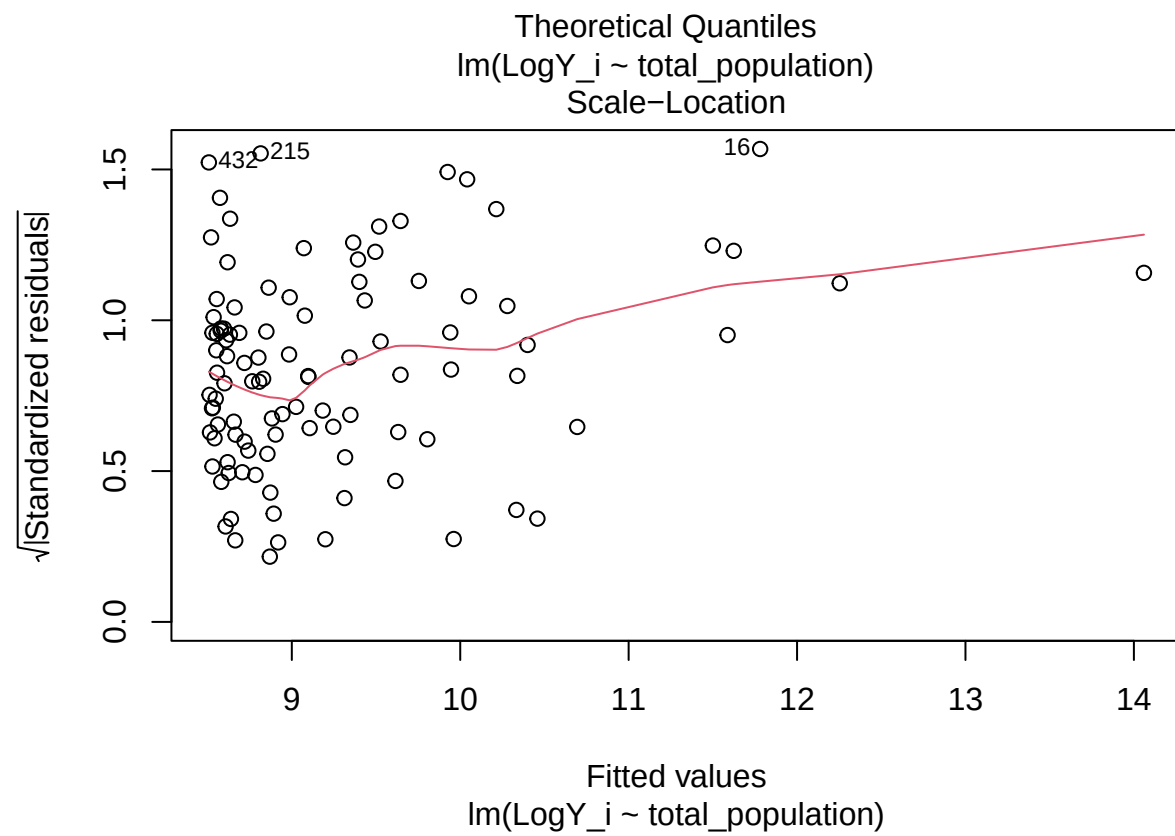
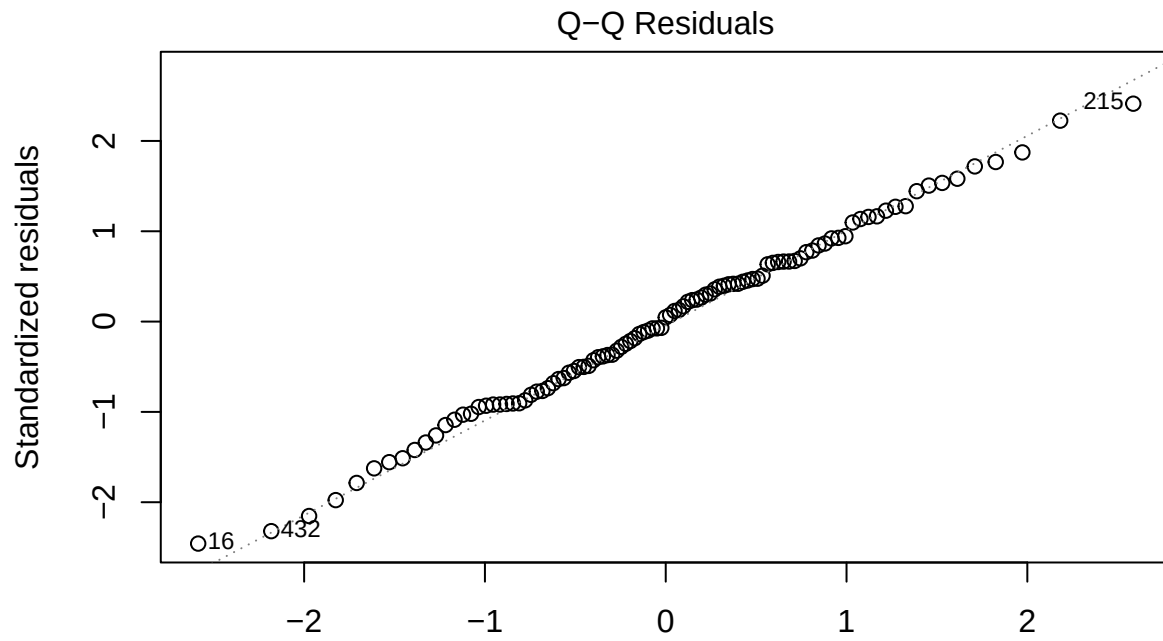
```
# REGION 1
summary(model_1) # all model stats
```

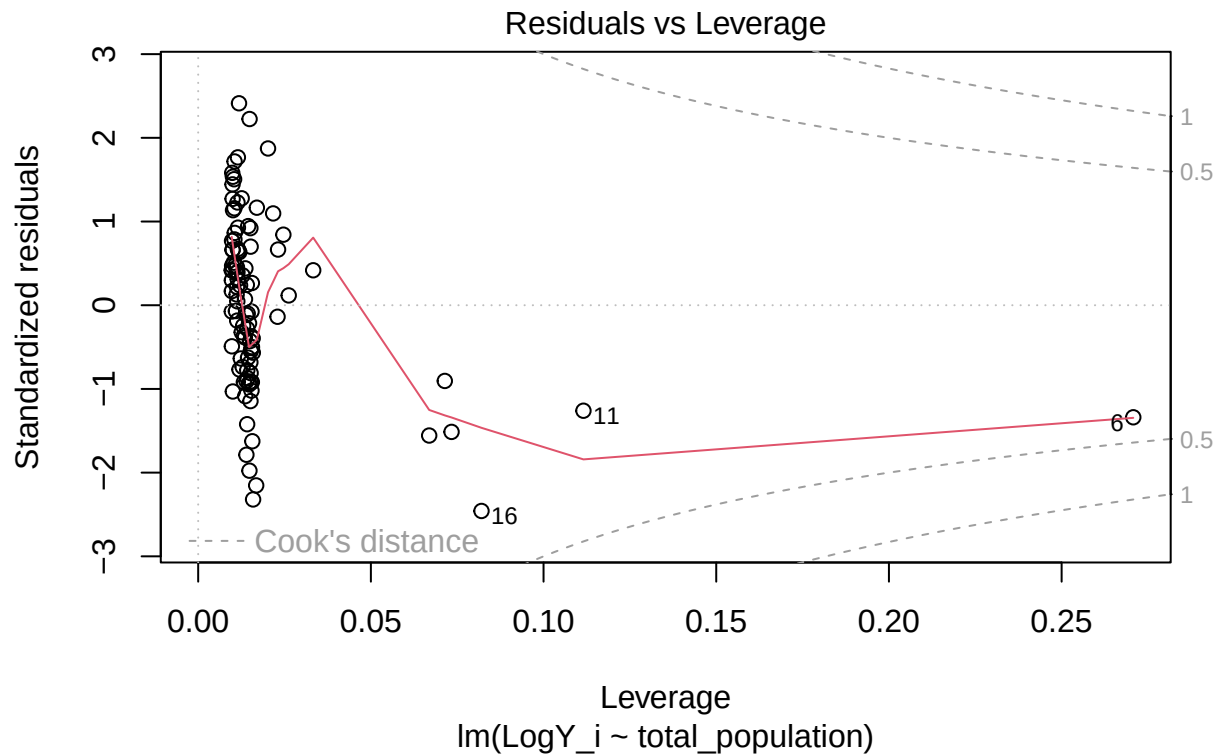
```
##
## Call:
## lm(formula = LogY_i ~ total_population, data = region1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -1.29446 -0.41112 0.02557 0.36235 1.31827
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.248e+00  7.962e-02  103.60  <2e-16 ***
## total_population 2.526e-06  1.474e-07   17.13  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5497 on 101 degrees of freedom
## Multiple R-squared:  0.744, Adjusted R-squared:  0.7415
## F-statistic: 293.5 on 1 and 101 DF, p-value: < 2.2e-16
```

```
plot(model_1) # all four diagnostic plots
```







```
confint(model_1) # 95% CI for both b0 and b1
```

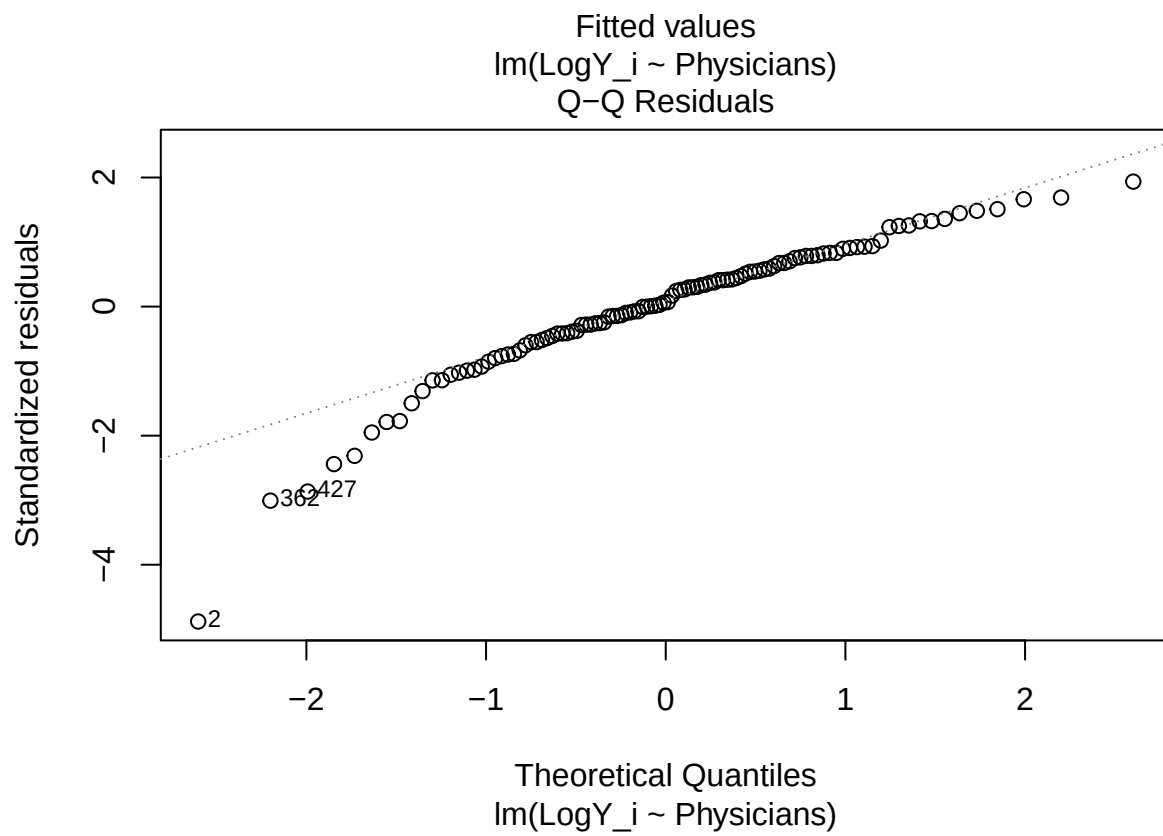
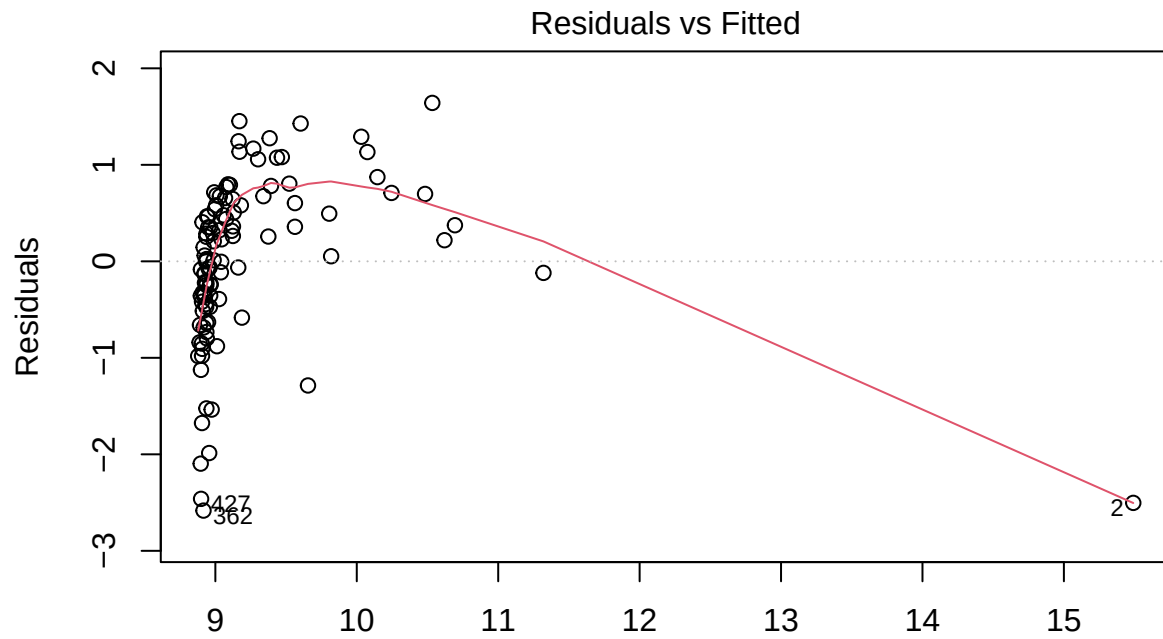
```
##                2.5 %      97.5 %
## (Intercept)    8.090439e+00 8.406322e+00
## total_population 2.233497e-06 2.818432e-06
```

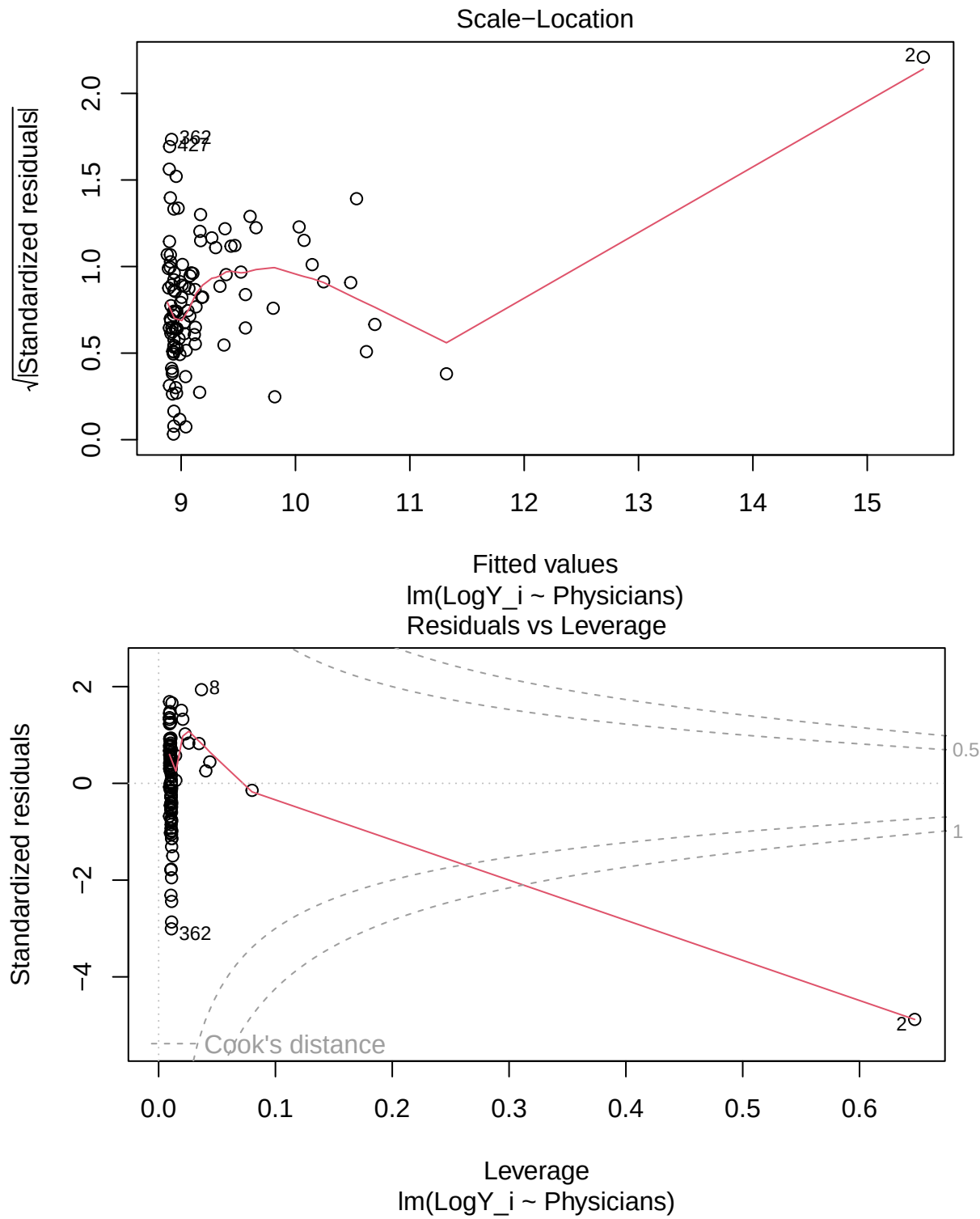
```
# REGION 2
```

```
summary(model_2)
```

```
##
## Call:
## lm(formula = LogY_i ~ Physicians, data = region2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.58290 -0.42535  0.05622  0.58644  1.64152
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.861e+00  9.290e-02  95.391  < 2e-16 ***
## Physicians   4.375e-04  4.826e-05   9.065  7.13e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8636 on 106 degrees of freedom
## Multiple R-squared:  0.4367, Adjusted R-squared:  0.4314
## F-statistic: 82.17 on 1 and 106 DF, p-value: 7.134e-15
```

```
plot(model_2) # insane outlier here
```





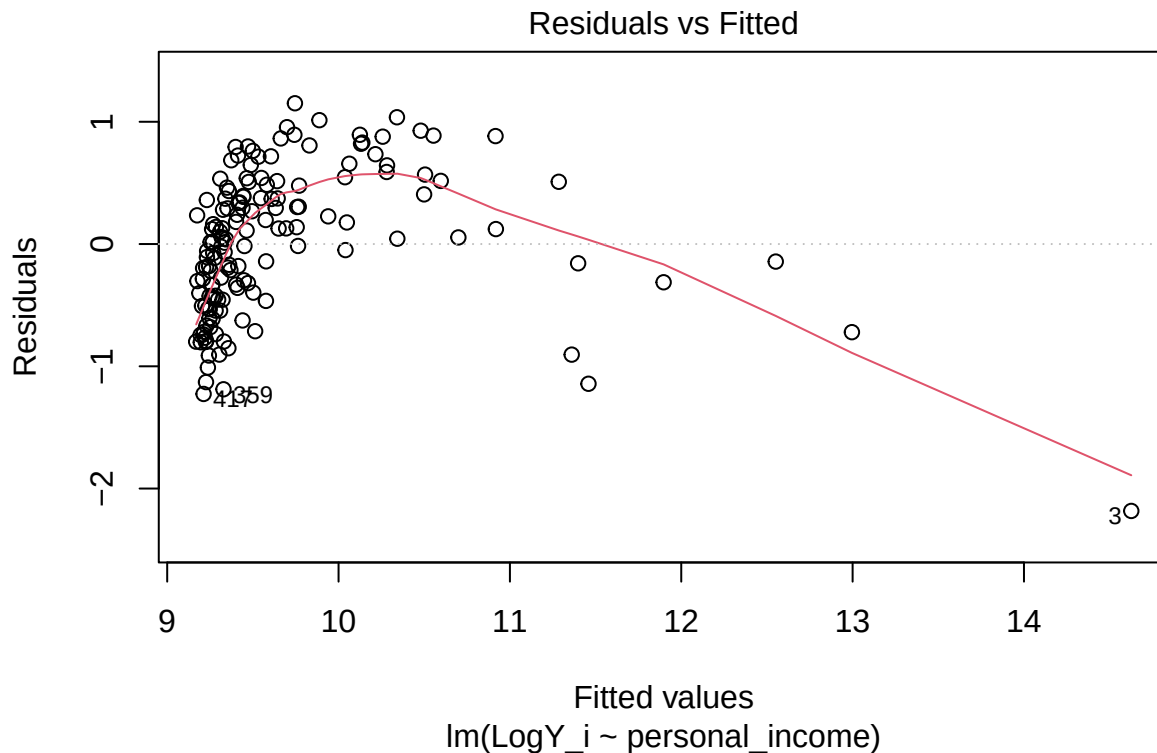
```
confint(model_2)
```

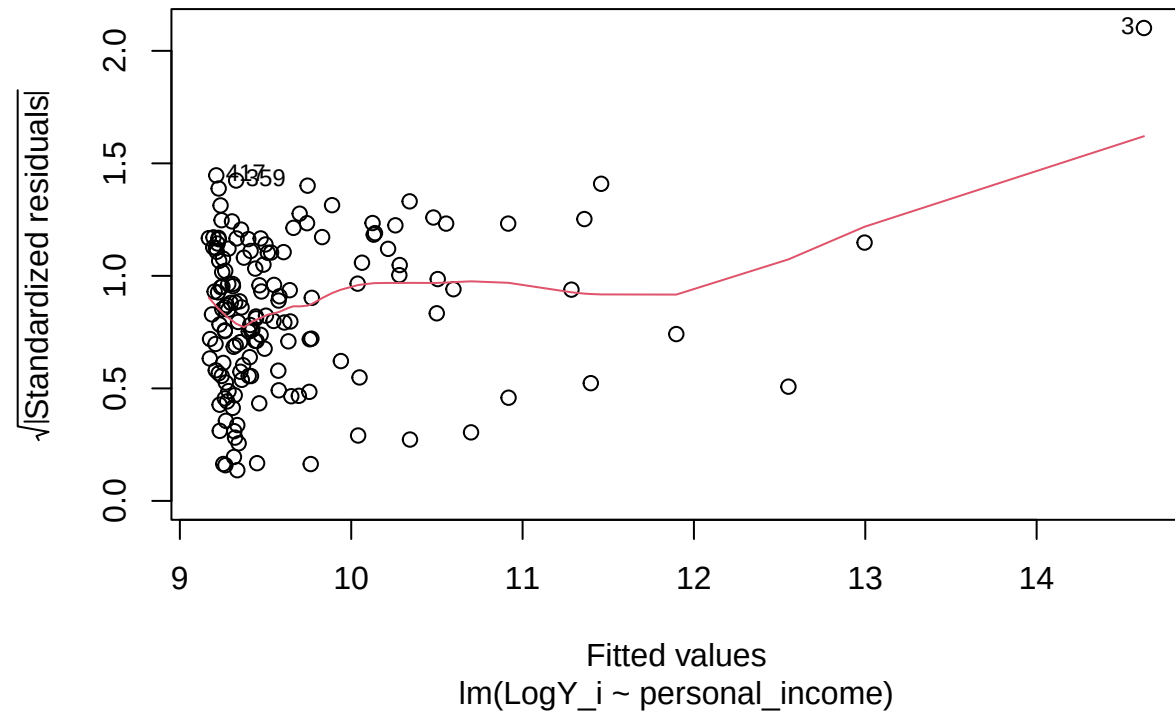
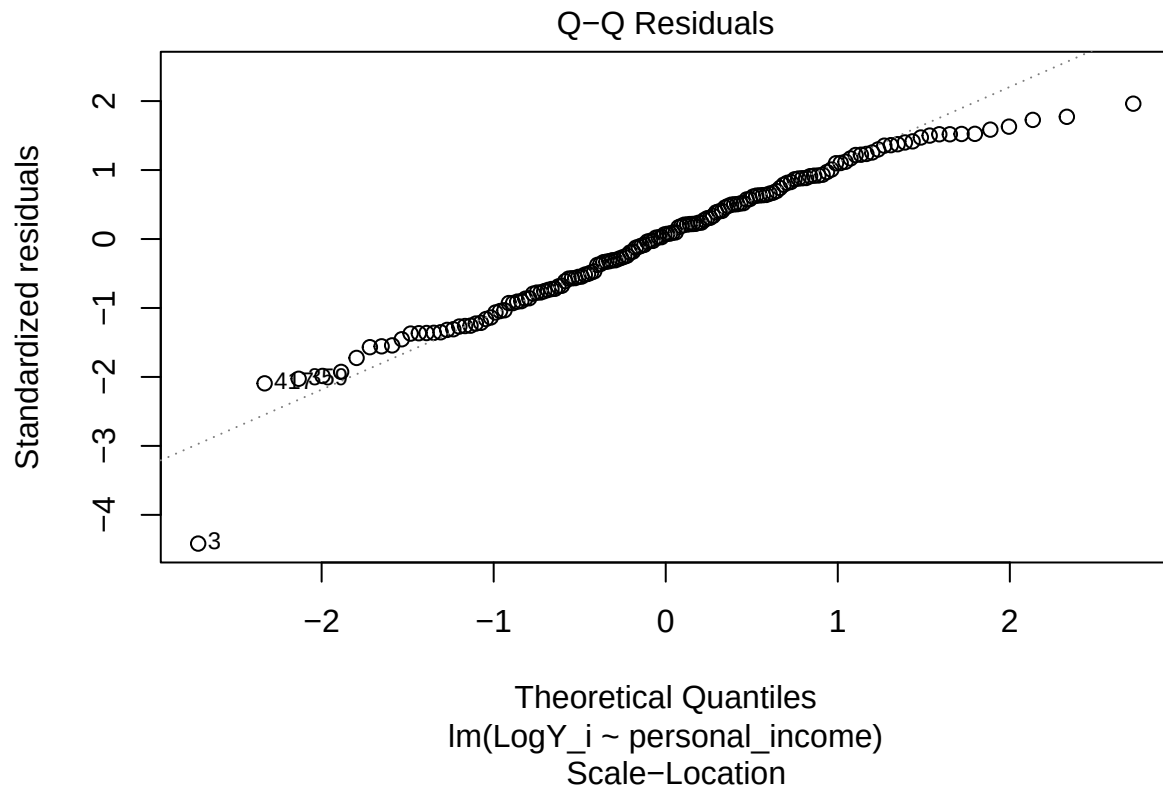
```
##              2.5 %      97.5 %
## (Intercept) 8.6773194178 9.0456722226
## Physicians  0.0003417957 0.0005331632
```

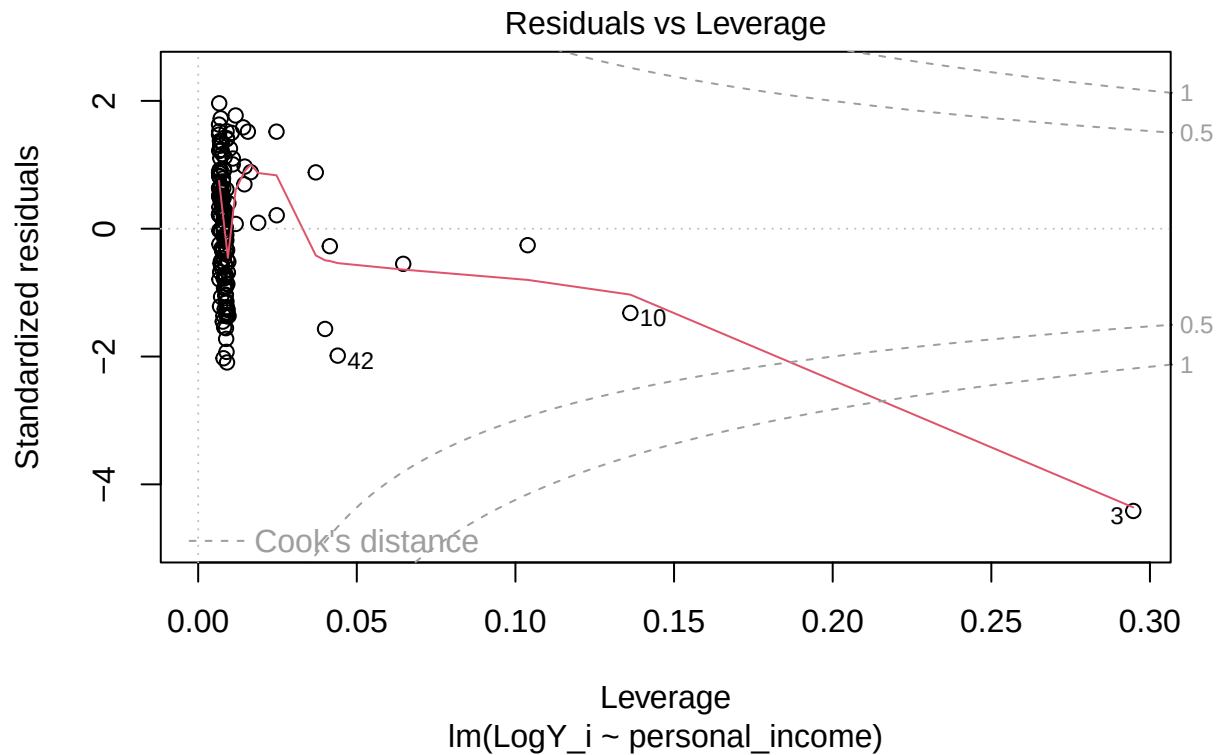
```
# REGION 3
summary(model_3)
```

```
##
## Call:
## lm(formula = LogY_i ~ personal_income, data = region3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.18304 -0.42726  0.04093  0.44053  1.15125
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.054e+00  6.205e-02  145.91  <2e-16 ***
## personal_income 1.013e-04  6.465e-06   15.67  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5886 on 150 degrees of freedom
## Multiple R-squared:  0.6208, Adjusted R-squared:  0.6183
## F-statistic: 245.6 on 1 and 150 DF,  p-value: < 2.2e-16
```

```
plot(model_3) # & here
```







```
confint(model_3)
```

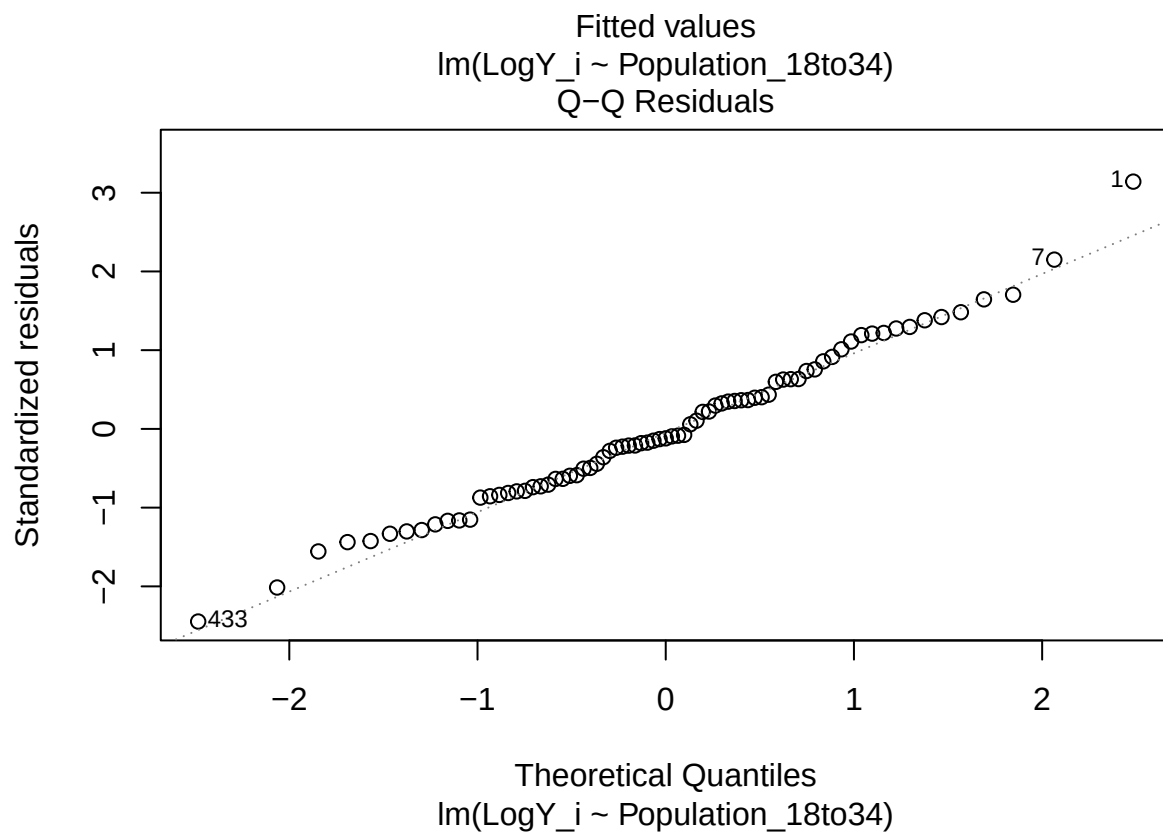
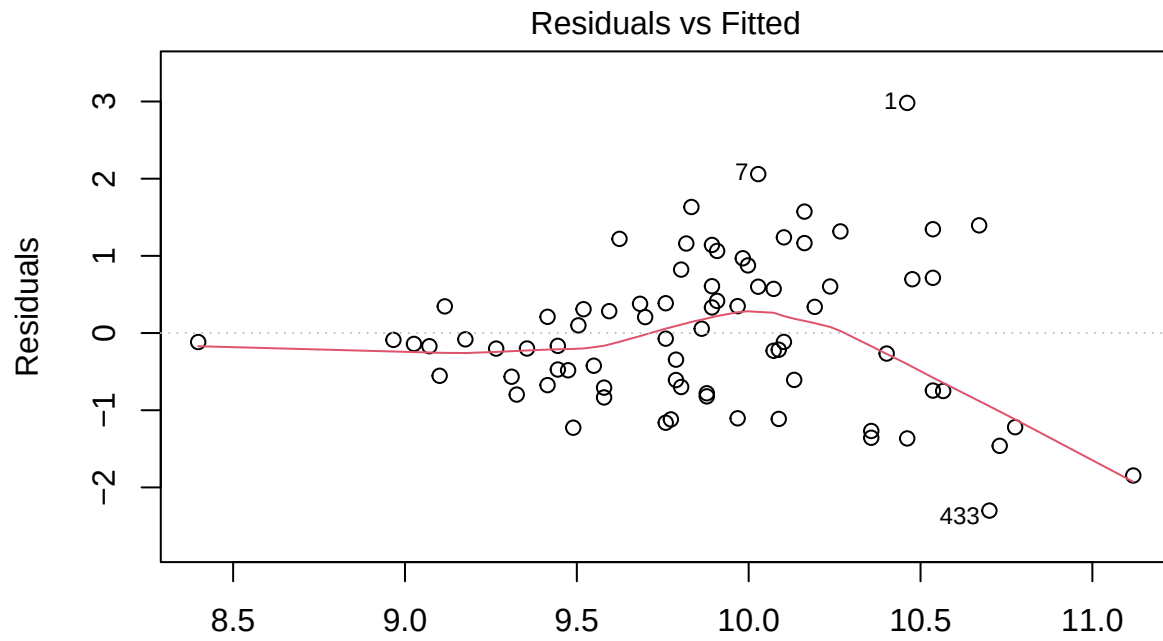
```
##                2.5 %      97.5 %
## (Intercept)    8.930933e+00 9.1761327786
## personal_income 8.854247e-05 0.0001140913
```

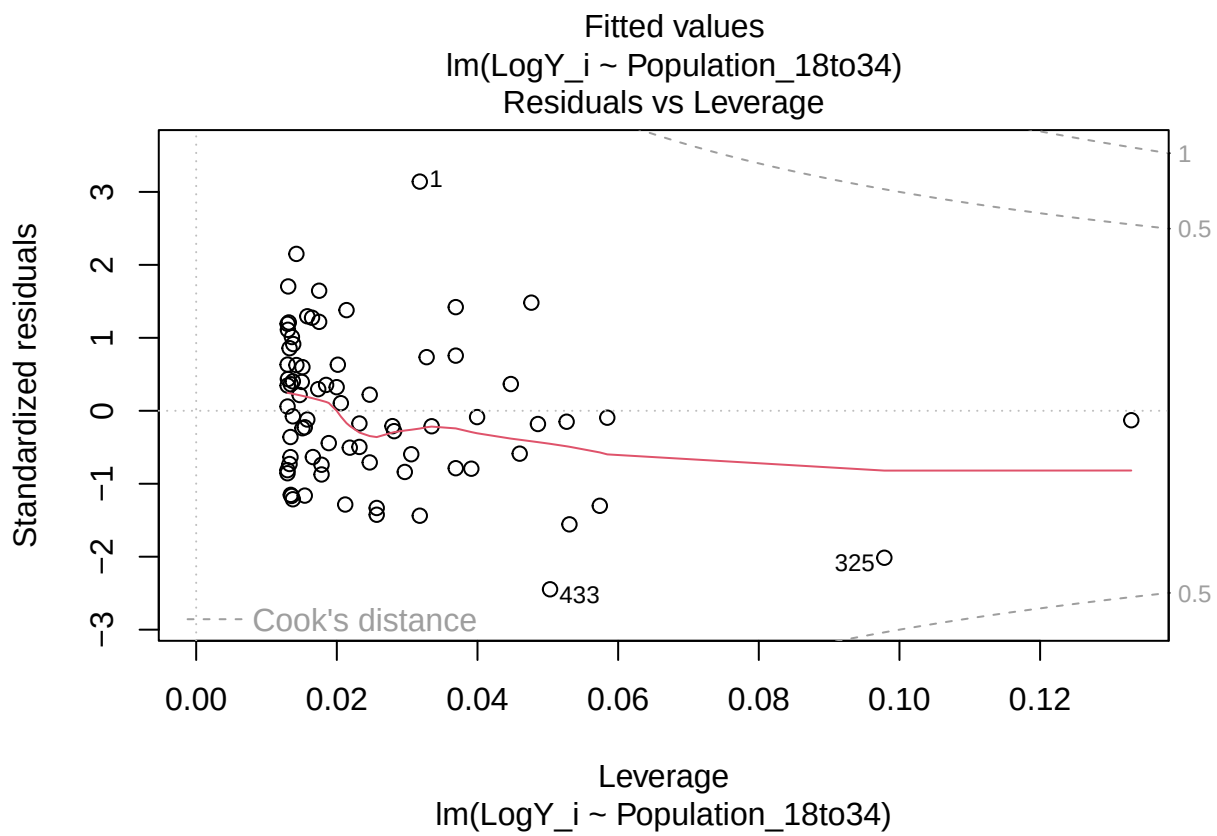
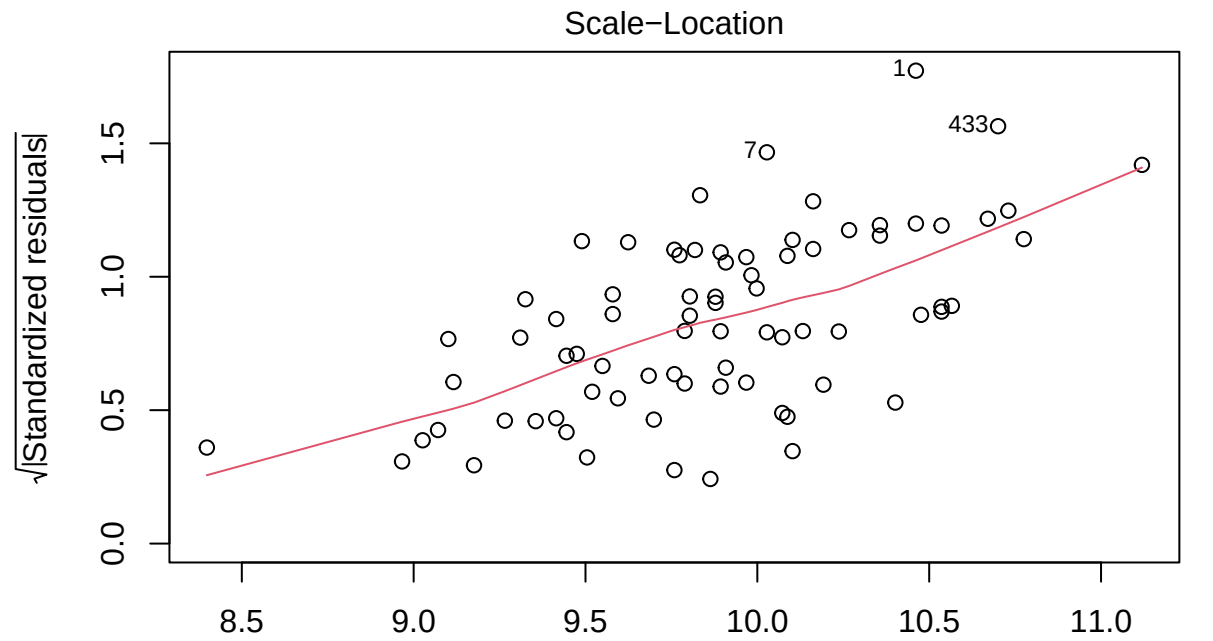
```
# REGION 4
```

```
summary(model_4)
```

```
##
## Call:
## lm(formula = LogY_i ~ Population_18to34, data = region4)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.3001 -0.6990 -0.1149  0.6031  2.9816
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.6626     0.9591   5.904 9.64e-08 ***
## Population_18to34  0.1495     0.0338   4.423 3.25e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9647 on 75 degrees of freedom
## Multiple R-squared:  0.2068, Adjusted R-squared:  0.1963
## F-statistic: 19.56 on 1 and 75 DF, p-value: 3.248e-05
```

```
plot(model_4)
```





```
confint(model_4)
```

```
##                2.5 %    97.5 %
## (Intercept)    3.75199556 7.5731234
## Population_18to34 0.08215539 0.2168313
```

REGION 1

For Region 1, we fit a simple linear regression of $\log(\text{total serious crimes})$ on the estimated total population. The estimated slope is 0.000002526 ($p = < 2.2\text{e-}16$ or ~ 0), suggesting a positive and significant relationship. The number is small despite the relationship being strong because total population is very large. The model explains 74.15% of the variation in $\log(\text{serious crimes})$.

The residual plots show a fanning pattern suggesting non-constant variance (heteroscedasticity), which motivates the transformation approach explored in Question 3. Q-Q looks good, residuals appear approximately normally distributed. There are multiple influential points pointed out in the Residuals v. Leverage plot (points 6, 11, & 16). Assumptions are not reasonably met.

We are 95% confident the intercept is between [8.09, 8.40] and that for each one unit increase in total population, $\log(Y)$ increases by between [0.0000022, 0.0000028]. Notably, the CI for β_1 does not include 0, which is evidence the predictor is significant.

REGION 2

For Region 2, we fit a simple linear regression of $\log(\text{total serious crimes})$ on the number of professionally active non-federal physicians. The estimated slope is 0.0004375 ($p = 7.134\text{e-}15$ or ~ 0), suggesting a positive and significant relationship. The model explains 43.14% of the variation in $\log(\text{serious crimes})$.

The residual plots show extreme fanning suggesting heteroscedasticity again. Points veer away from the ends of the Q-Q plot line, so residuals are not entirely normally distributed. There is an extremely influential outlier (point 2) pointed out in the Residuals v. Leverage plot that greatly exceeds Cook's distance and is likely pulling hard on the regression line. Assumptions are not reasonably met.

We are 95% confident the intercept is between [8.67, 9.04] and that for each physician, $\log(Y)$ increases by between [0.0003, 0.0005]. Notably, the CI for β_1 does not include 0, which is evidence the predictor is significant.

REGION 3

For Region 3, we fit a simple linear regression of $\log(\text{total serious crimes})$ on the total personal income of CDI population (in millions of dollars). The estimated slope is 0.0001013 ($p = < 2.2\text{e-}16$ or ~ 0), suggesting a positive and significant relationship. The model explains 61.83% of the variation in $\log(\text{serious crimes})$.

Once again, residual plots show extreme fanning suggesting heteroscedasticity. Points veer only slightly away from Q-Q plot, residuals appear approximately normally distributed. Again, there is an extremely influential outlier (point 3) that greatly exceeds Cook's distance. Assumptions are not reasonably met.

We are 95% confident the intercept is between [8.93, 9.17] and that for each one unit increase in personal income, $\log(Y)$ increases by between [0.000088, 0.00011]. Notably, the CI for β_1 does not include 0, which is evidence the predictor is significant.

REGION 4

For Region 4, we fit a simple linear regression of $\log(\text{total serious crimes})$ on percent of CDI population aged 18-34. The estimated slope is 0.1495 ($p = 3.248\text{e-}05$, quite small), suggesting a positive and significant relationship. The model explains 19.63% of the variation in $\log(\text{serious crimes})$, which is quite low! This predictor is significant but not strong.

Residual plot is approximately randomly distributed, suggesting mostly constant variance. Q-Q points closely follow line. Some influential points (points 433 and 325) come close to Cook's distance cutoff but don't exceed it. Assumptions are reasonably met!

We are 95% confident the intercept is between [3.75, 7.57] and that for each one unit increase in population of 18-34 year olds, $\log(Y)$ increases by between [0.082, 0.21]. Notably, the CI for β_1 does not include 0, which is evidence the predictor is significant.

Question 3 : Model Refinement Using Transformations

```
# REGION 1
model_1b <- lm(log(Demographic$Y_i) ~ log(Demographic$total_population), data = region1)

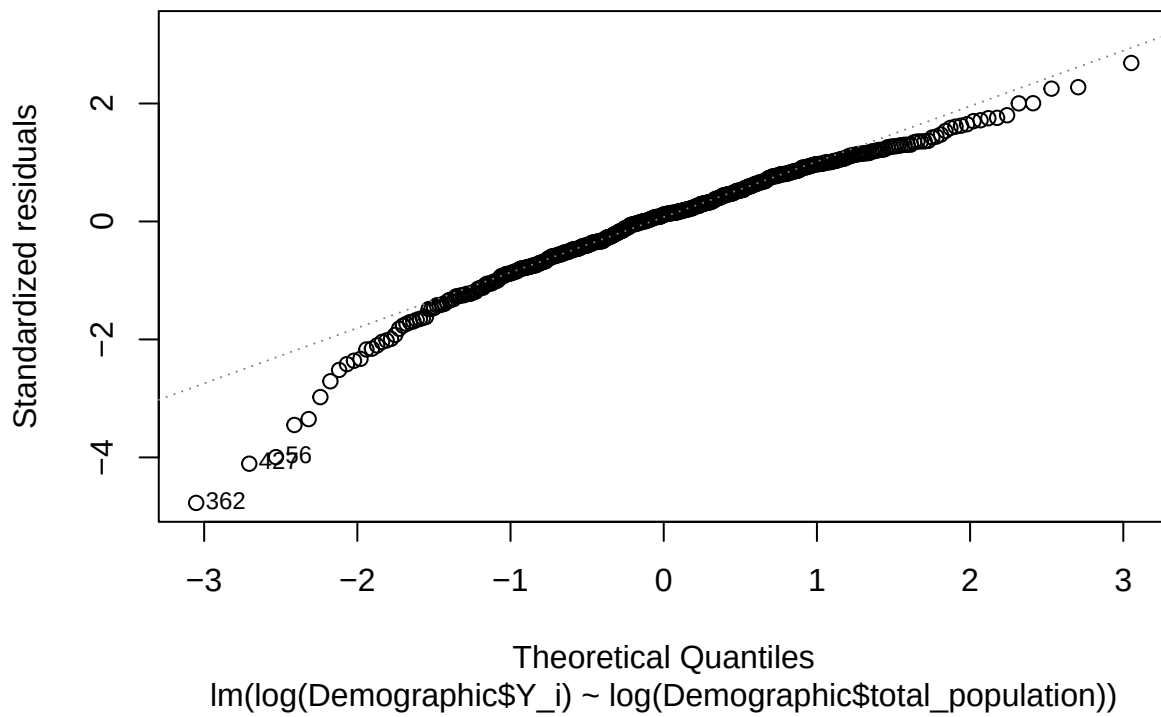
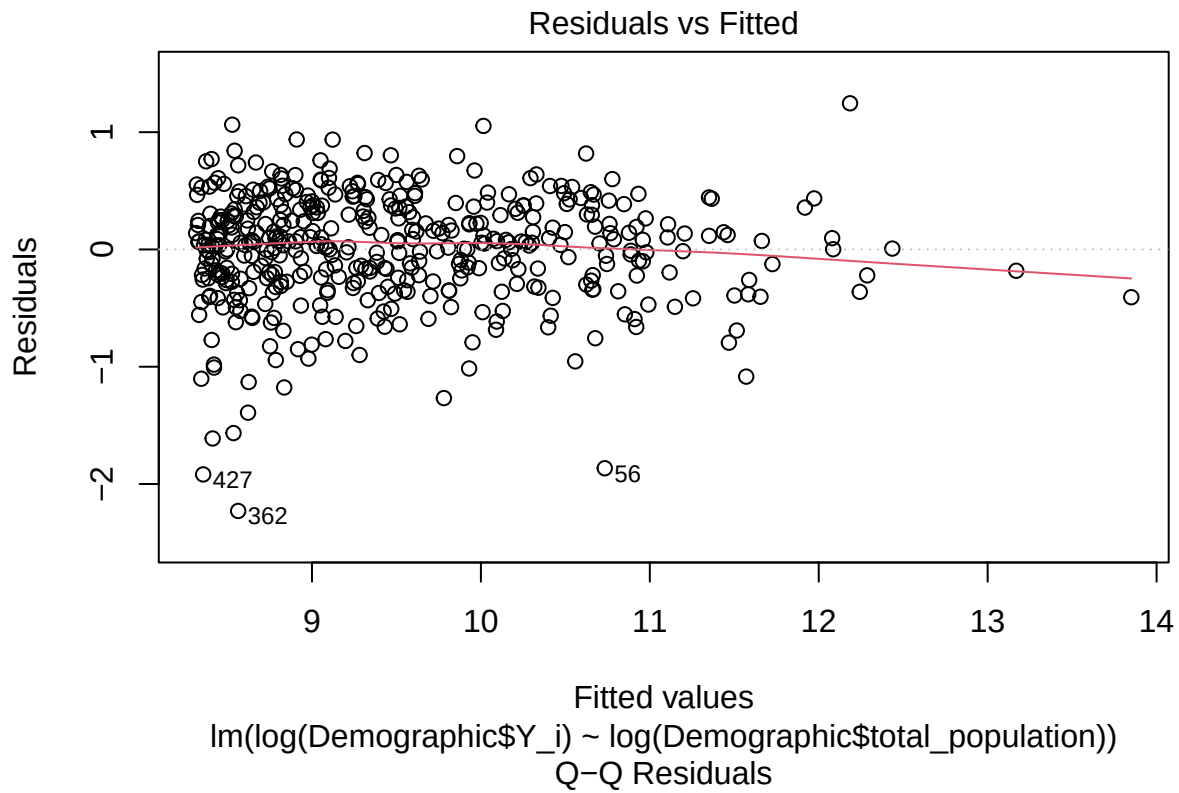
anova(model_1b)
```

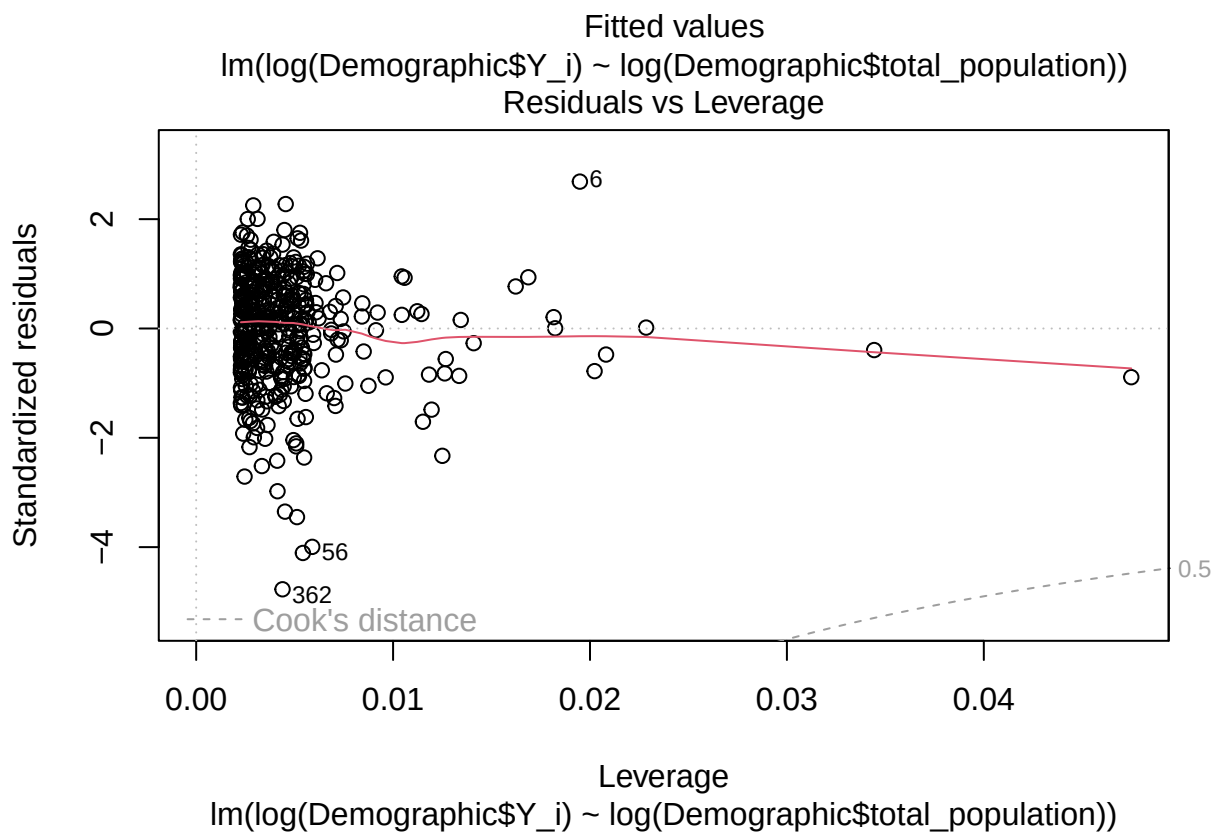
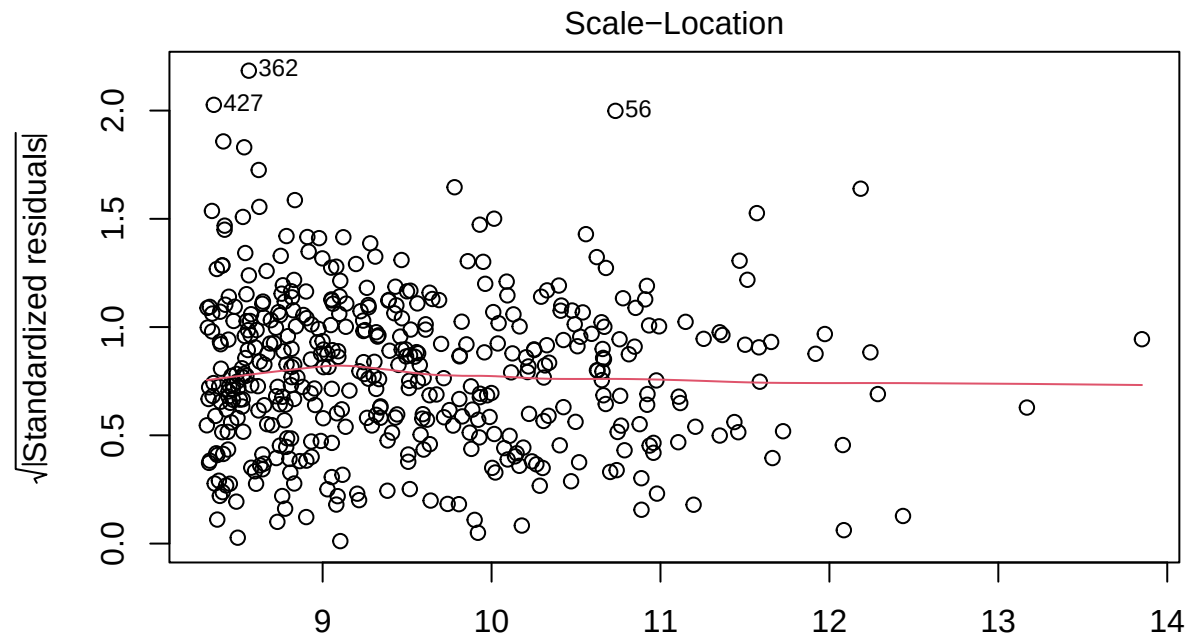
```
## Analysis of Variance Table
##
## Response: log(Demographic$Y_i)
##
##              Df Sum Sq Mean Sq F value    Pr(>F)
## log(Demographic$total_population)    1 417.98   417.98  1905.3 < 2.2e-16 ***
## Residuals                        438  96.09     0.22
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model_1b)
```

```
##
## Call:
## lm(formula = log(Demographic$Y_i) ~ log(Demographic$total_population),
##     data = region1)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.22974 -0.26043  0.05404  0.33178  1.24605
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -5.89928     0.35357  -16.68  <2e-16 ***
## log(Demographic$total_population)  1.23455     0.02828   43.65  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4684 on 438 degrees of freedom
## Multiple R-squared:  0.8131, Adjusted R-squared:  0.8127
## F-statistic: 1905 on 1 and 438 DF, p-value: < 2.2e-16
```

```
plot(model_1b) # greatly improved linearity
```





```
confint(model_1b)
```

```
##                2.5 %    97.5 %
## (Intercept)    -6.594179 -5.204383
## log(Demographic$total_population)  1.178958  1.290133
```



```
# REGION 2
model_2b <- lm(log(Demographic$Y_i) ~ log(Demographic$Physicians), data=region2)

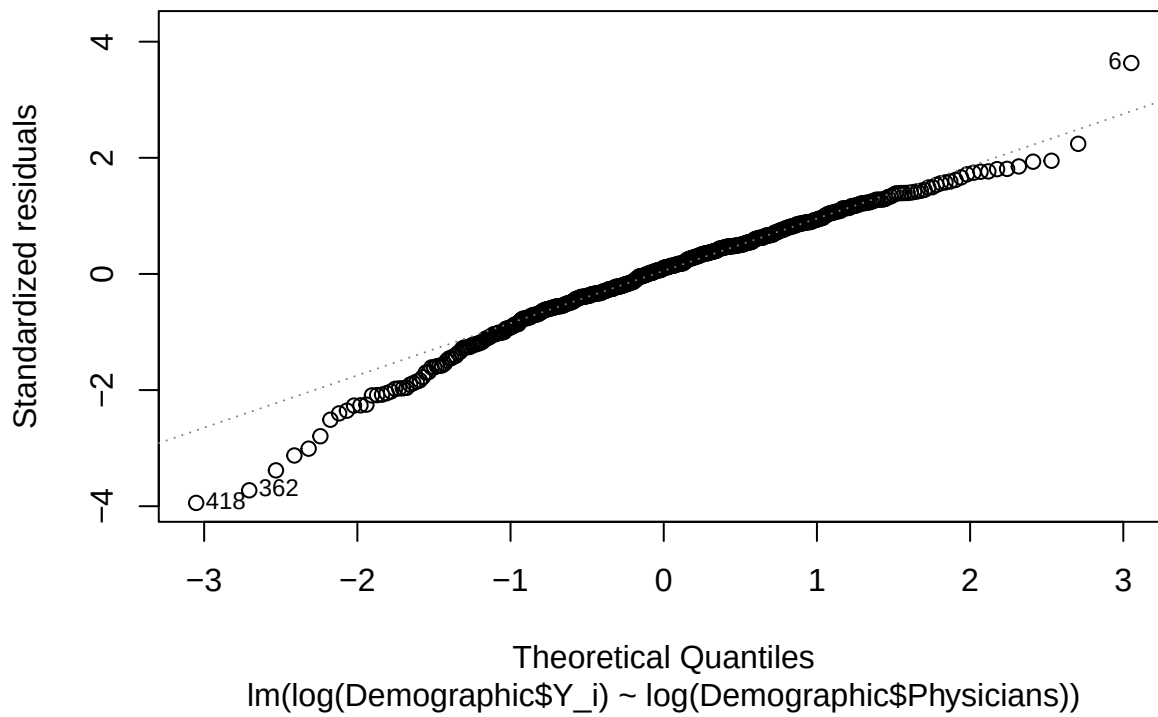
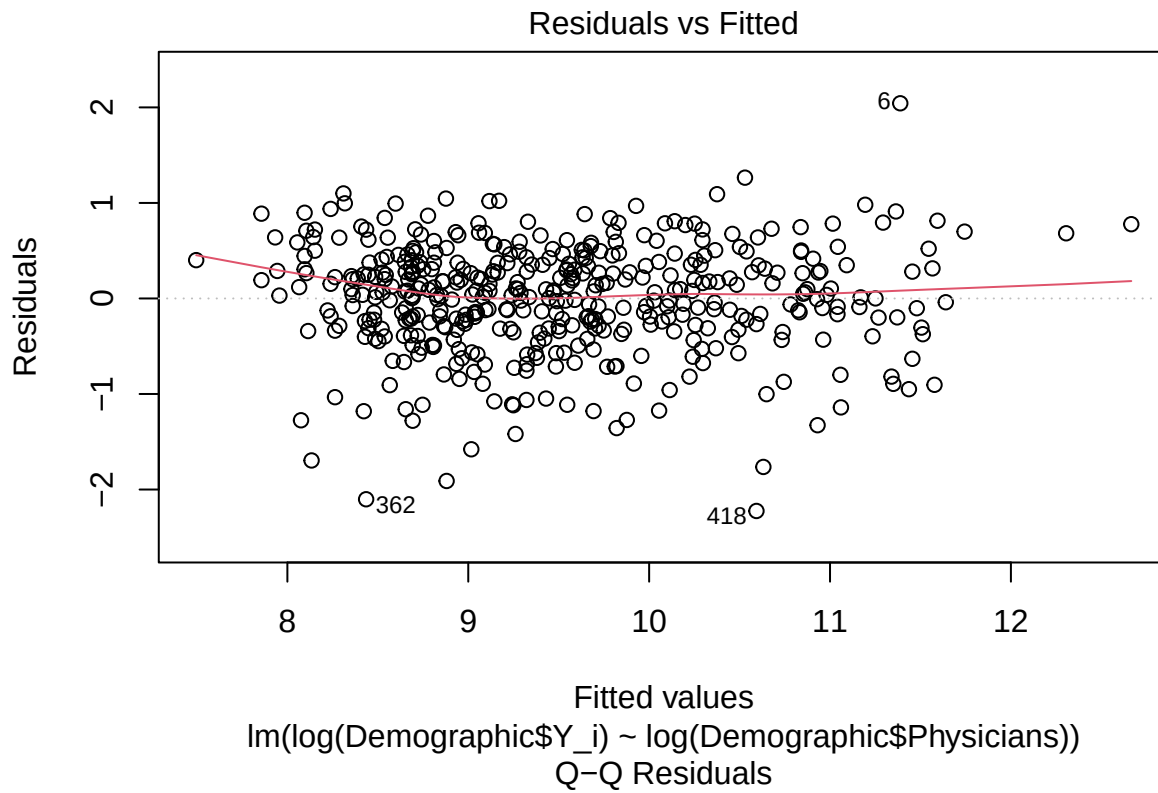
anova(model_2b)
```

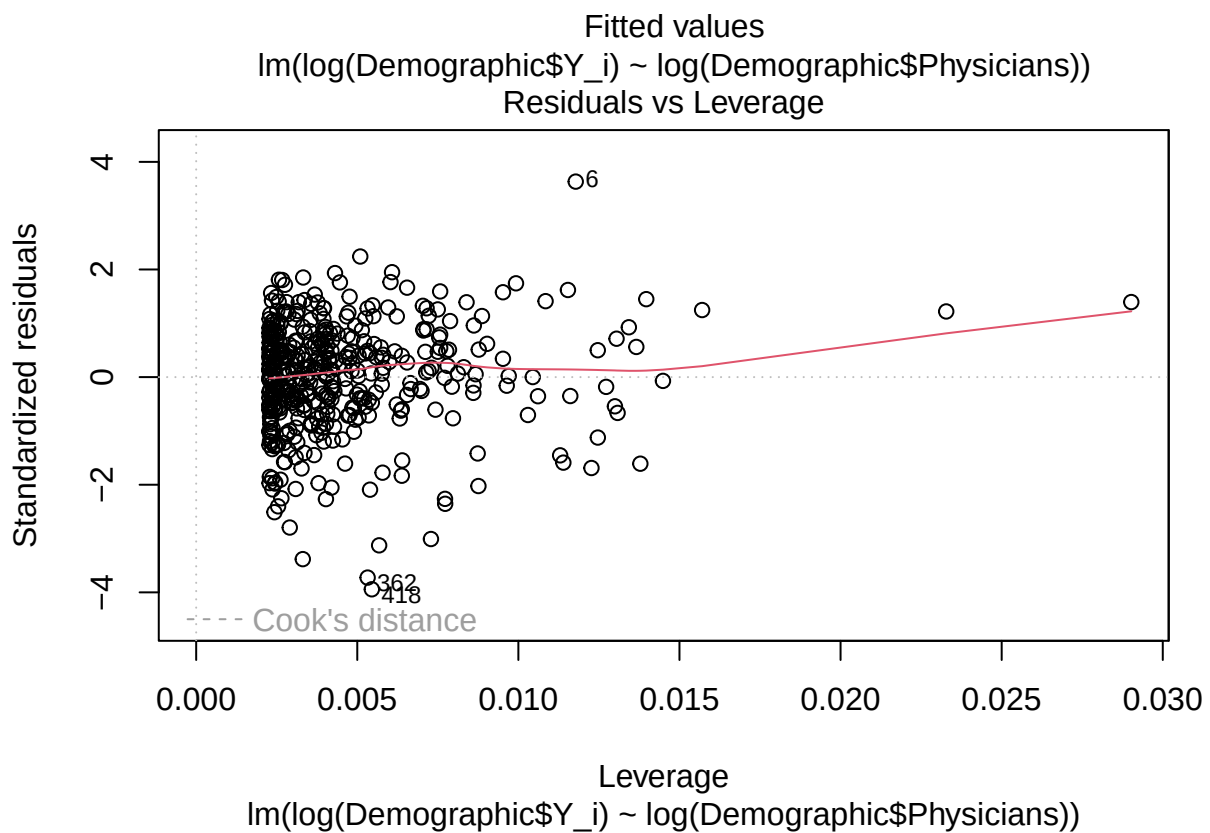
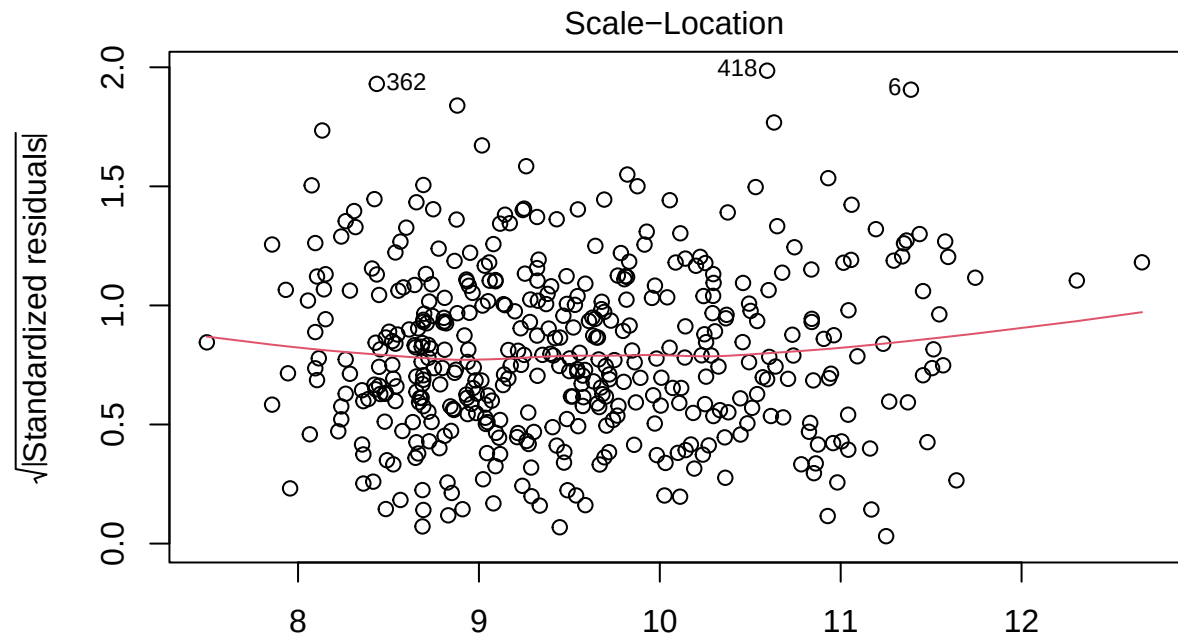
```
## Analysis of Variance Table
##
## Response: log(Demographic$Y_i)
##              Df Sum Sq Mean Sq F value    Pr(>F)
## log(Demographic$Physicians)    1 373.90   373.90  1168.4 < 2.2e-16 ***
## Residuals                    438 140.17    0.32
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model_2b)
```

```
##
## Call:
## lm(formula = log(Demographic$Y_i) ~ log(Demographic$Physicians),
##     data = region2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.22444 -0.31244  0.05857  0.37353  2.04298
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      4.5404     0.1477   30.75 <2e-16 ***
## log(Demographic$Physicians)  0.8067     0.0236   34.18 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5657 on 438 degrees of freedom
## Multiple R-squared:  0.7273, Adjusted R-squared:  0.7267
## F-statistic: 1168 on 1 and 438 DF, p-value: < 2.2e-16
```

```
plot(model_2b)
```





```
confint(model_2b)
```

```
##              2.5 %    97.5 %
## (Intercept)    4.2501706 4.8306031
## log(Demographic$Physicians) 0.7602963 0.8530617
```

```
# REGION 3
model_3b <- lm(log(Demographic$Y_i) ~ log(Demographic$personal_income), data = region3)

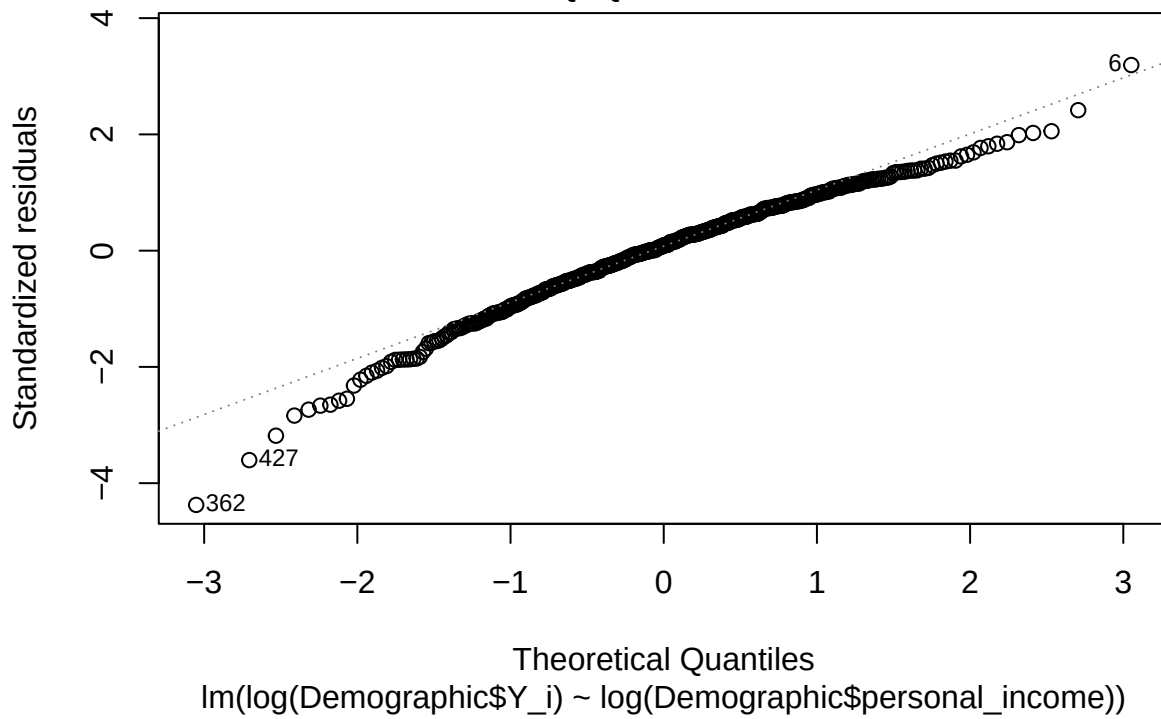
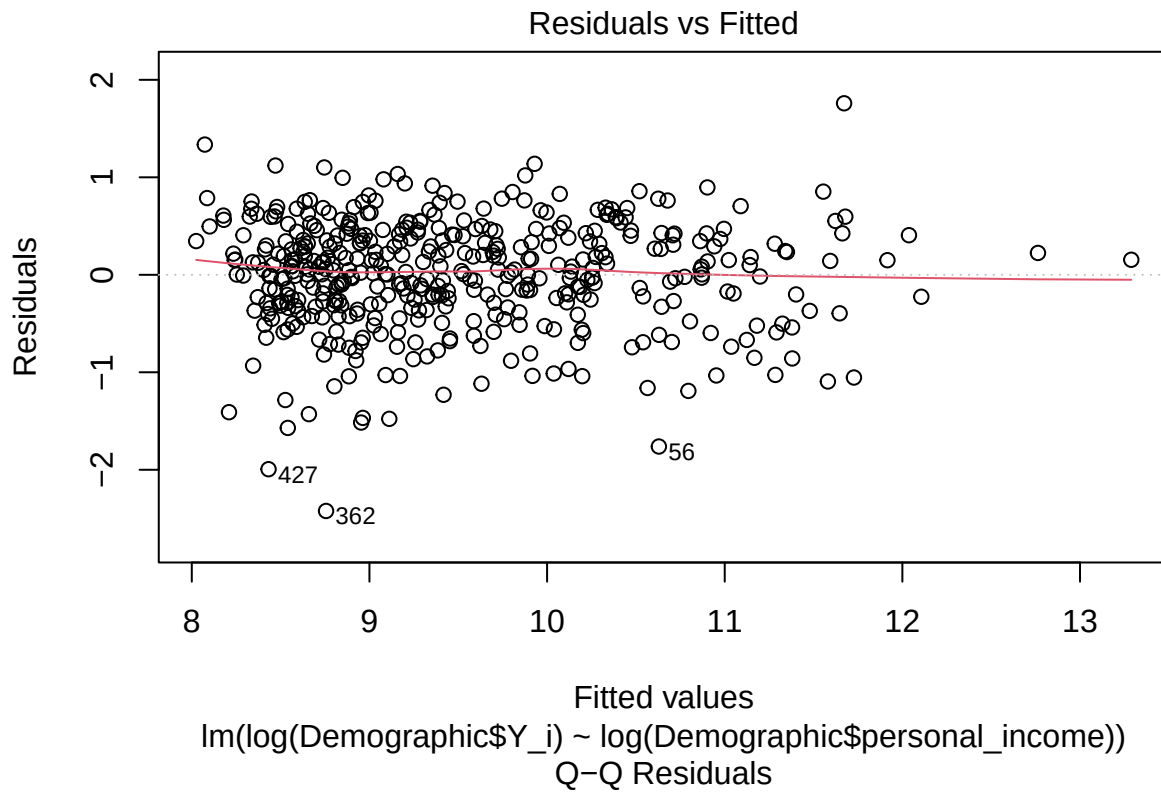
anova(model_3b)
```

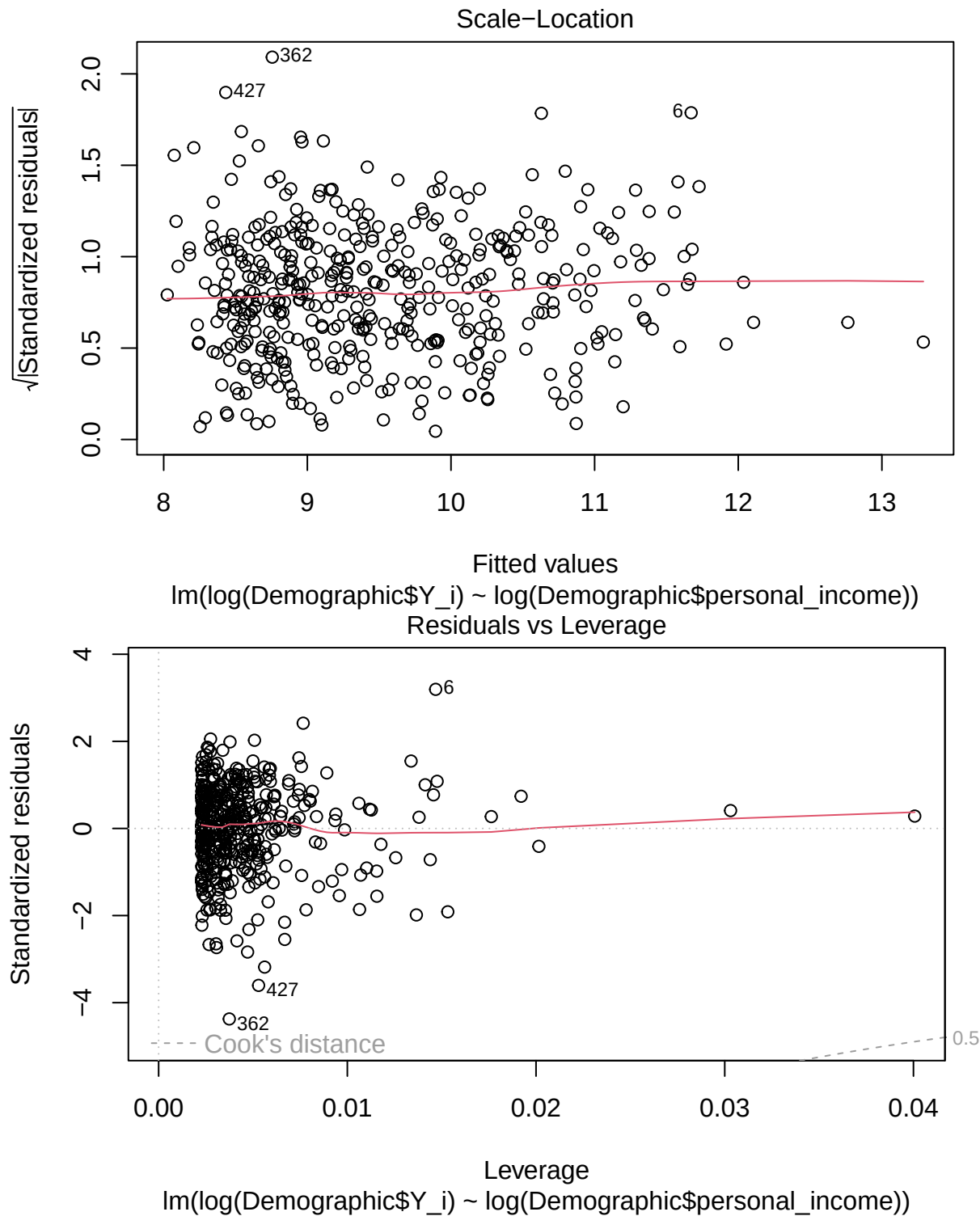
```
## Analysis of Variance Table
##
## Response: log(Demographic$Y_i)
##
##           Df Sum Sq Mean Sq F value    Pr(>F)
## log(Demographic$personal_income)  1 379.20   379.20  1231.5 < 2.2e-16 ***
## Residuals                        438 134.87    0.31
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model_3b)
```

```
##
## Call:
## lm(formula = log(Demographic$Y_i) ~ log(Demographic$personal_income),
##     data = region3)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.42255 -0.31757  0.04828  0.40349  1.75927
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.7363     0.2512   2.931  0.00356 **
## log(Demographic$personal_income)  1.0353     0.0295  35.092 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5549 on 438 degrees of freedom
## Multiple R-squared:  0.7376, Adjusted R-squared:  0.737
## F-statistic: 1231 on 1 and 438 DF, p-value: < 2.2e-16
```

```
plot(model_3b) # greatly improved linearity, previous outliers arent as bad now
```





```
confint(model_3b)
```

```
##                2.5 %   97.5 %
## (Intercept)    0.2425550 1.230022
## log(Demographic$personal_income) 0.9773465 1.093318
```

```
# REGION 4
model_4b <- lm(log(Demographic$Y_i) ~ log(Demographic$Population_18to34), data = region4)

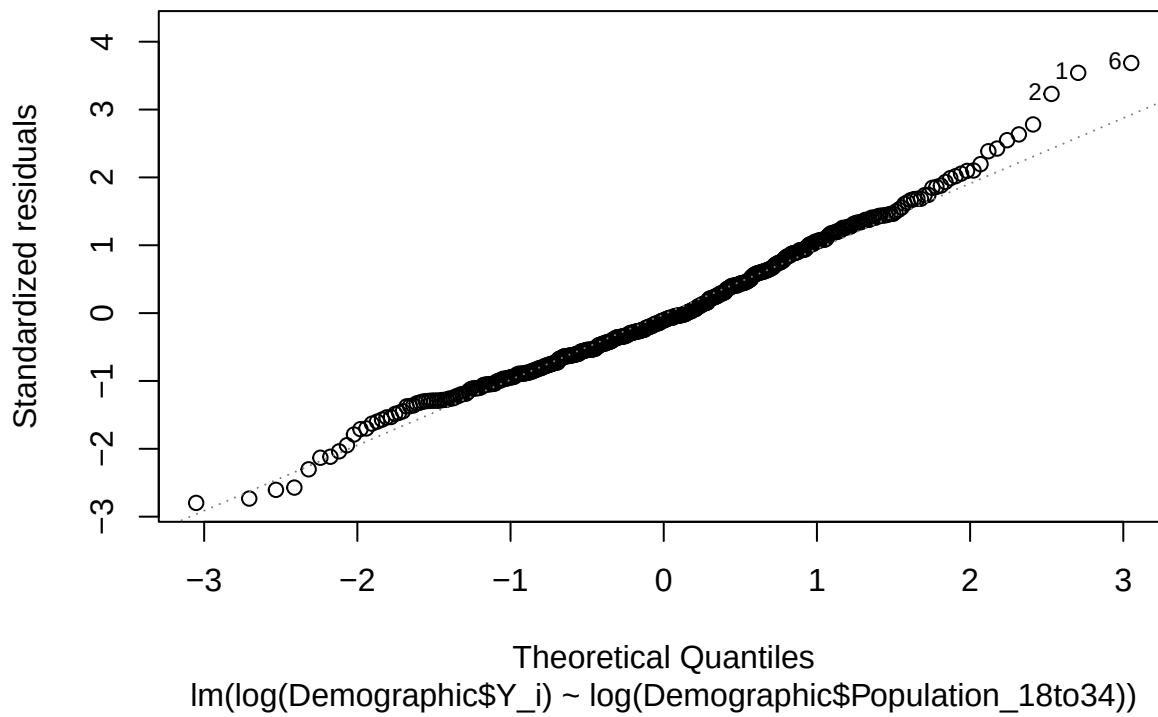
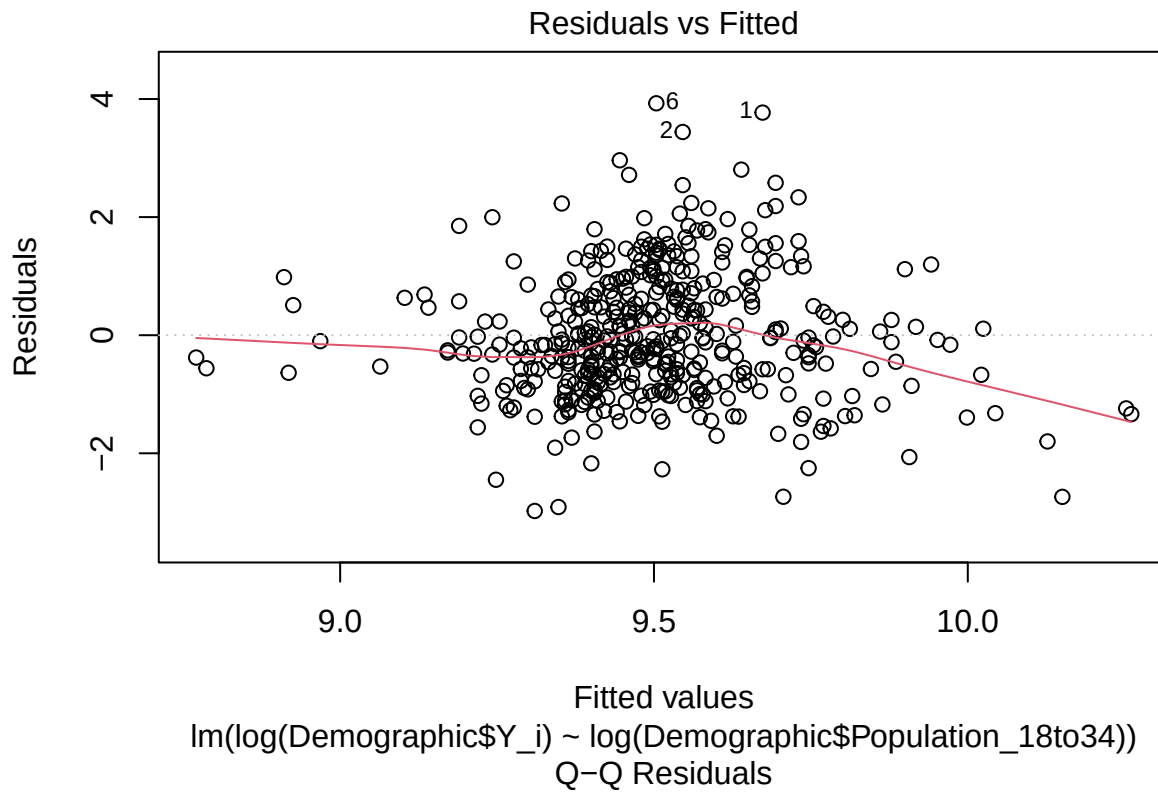
anova(model_4b)
```

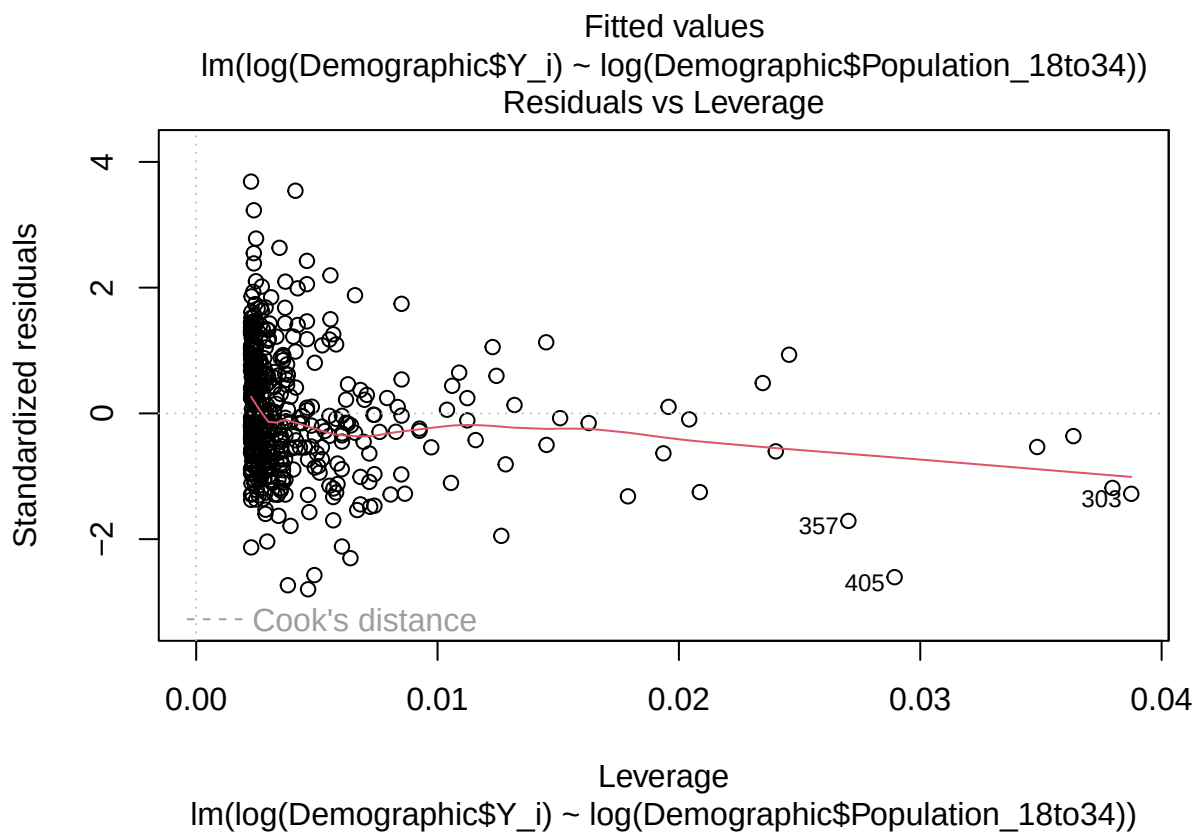
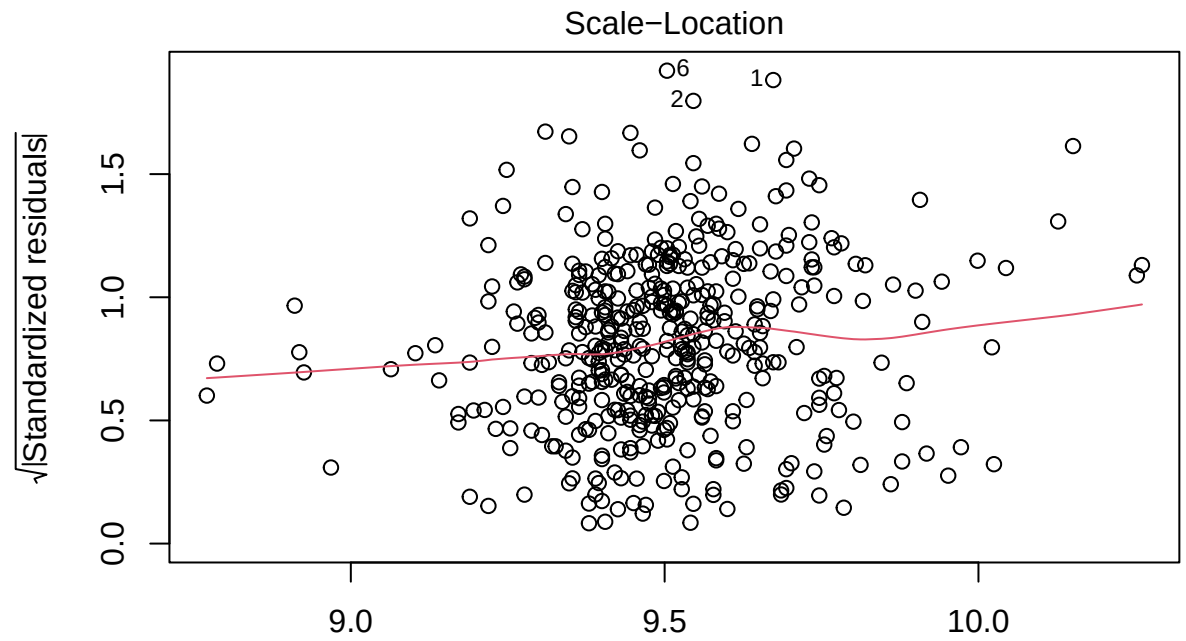
```
## Analysis of Variance Table
##
## Response: log(Demographic$Y_i)
##
##              Df Sum Sq Mean Sq F value    Pr(>F)
## log(Demographic$Population_18to34)    1   15.74   15.7386   13.833 0.0002257 ***
## Residuals                        438  498.33    1.1377
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model_4b)
```

```
##
## Call:
## lm(formula = log(Demographic$Y_i) ~ log(Demographic$Population_18to34),
##     data = region4)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.9767 -0.7128 -0.1149  0.6731  3.9275
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      5.0116     1.2086   4.147 4.05e-05 ***
## log(Demographic$Population_18to34)  1.3438     0.3613   3.719 0.000226 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.067 on 438 degrees of freedom
## Multiple R-squared:  0.03062,    Adjusted R-squared:  0.0284
## F-statistic: 13.83 on 1 and 438 DF,  p-value: 0.0002257
```

```
plot(model_4b)
```





```
confint(model_4b)
```

```
##                2.5 %   97.5 %
## (Intercept)    2.6361888 7.387033
## log(Demographic$Population_18to34) 0.6336964 2.053905
```

REGION 1

For Region 1, we applied a log transformation onto the predictor (estimated total population). The estimated slope is 1.234, suggesting a positive and significant relationship. The model now explains 81.27% ($R^2 = 0.8127$) of the variation in $\log(\text{serious crimes})$, an increase from the previous 74.15%.

Compared to the original model, the residuals plots of the transformed model display a drastic increase in homoscedastic residuals. However, when plotting the residuals, the data points appear to follow a slight funnel shape, though this does not appear to be dramatic enough to constitute an egregious violation of assumptions. On the Q-Q plot, the points still mostly form a straight line, with points only veering off slightly near the ends. The residuals appear approximately normally distributed just as it was with the original model. In regards to the Residuals vs. Leverage plot, the previously influential points (points 6, 11, & 16) of the original model are no longer as influential in the transformed model. Assumptions are now reasonably met. The model overall appears to be a better fit for the data after the transformation and the quality of inference seems to have dramatically improved for these reasons.

We are 95% confident the intercept is between $[-6.594, -5.204]$ and that for each one unit increase in personal income, $\log(Y)$ increases by between $[1.178, 1.290]$. The confidence intervals for both β_0 and β_1 became much wider. Notably, the CI for β_1 does not include 0, which is evidence the predictor is significant. The p-values for β_0 and β_1 remained the same at $2e^{-16}$ after the transformation.

REGION 2

For Region 2, we applied the log transformation to the original model, the slope of the model is approximately 0.806 ($p = < 2e^{-16}$ or ~ 0), suggesting a positive and significant relationship. The p-value for the transformed model is slightly smaller than the original model however both are approximately zero. The transformed CI $[0.760, 0.853]$ is not smaller than the original model, while this initially may indicate that the original model is better. However with further analysis of the Residuals vs. Fitted we can see significant improvement with the transformed data where the data is showing homoscedasticity in the Residuals vs. Fitted plot. We can also see in the Residuals vs. Leverage plot that the mentioned outlier is no longer present and is better fitting. The Q-Q Residuals plot also shows the residuals being approximately normally distributed but not much variation from the original model in this plot. The R^2 value is higher for the transformed data original ($R^2 = 0.4314$), transformed ($R^2 = 0.7267$), we can see a large increase in the model variability being explained by the predictor variable which is physicians in this case. Based off the improvement in the data matching the transformed model we will choose the log transformation to represent our model for region 2.

REGION 3

For Region 3, we applied a log transformation onto the predictor (personal income of CDI population (in millions of dollars)). The estimated slope is 1.035, suggesting a positive and significant relationship. The model now explains 73.7% ($R^2 = 0.737$) of the variation in $\log(\text{serious crimes})$, an increase from the previous 61.83%.

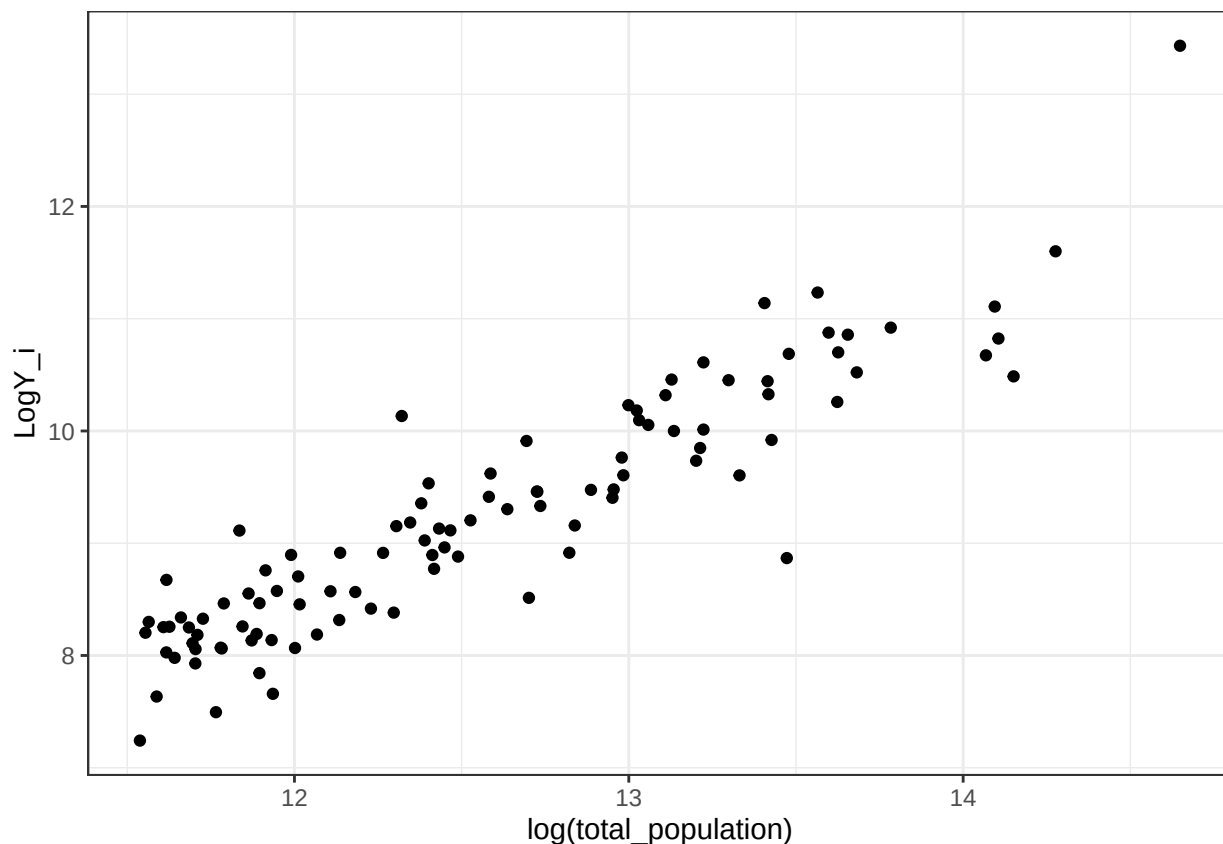
Contrary to the original model, the residuals plot of the transformed model displays homoscedasticity. On the Q-Q plot, the points still mostly form a straight line, with points only veering off slightly near the ends. The residuals appear approximately normally distributed just as it was with the original model. The previous outlier in the original model (point 3) no longer exceeds Cook's Distance and no longer appears to be an outlier relative to the data. Assumptions are now reasonably met. The model overall appears to be a better fit for the data after the transformation and the quality of inference seems to have dramatically improved for these reasons.

We are 95% confident the intercept is between $[0.242, 1.23]$ and that for each one unit increase in personal income, $\log(Y)$ increases by between $[0.977, 1.093]$. The confidence intervals for both β_0 and β_1 became much wider. Notably, the CI for β_1 does not include 0, which is evidence the predictor is significant. The p-value for β_0 increased from $2e^{-16}$ to 0.00356, while the p-value for β_1 remained the same at $2.2e^{-16}$ after the transformation.

REGION 4 For Region 4, we applied the log transformation to the original model, the slope of the model is approximately 1.343 ($p = 2.26e^{-04}$), suggesting a positive and significant relationship. The p-value for the

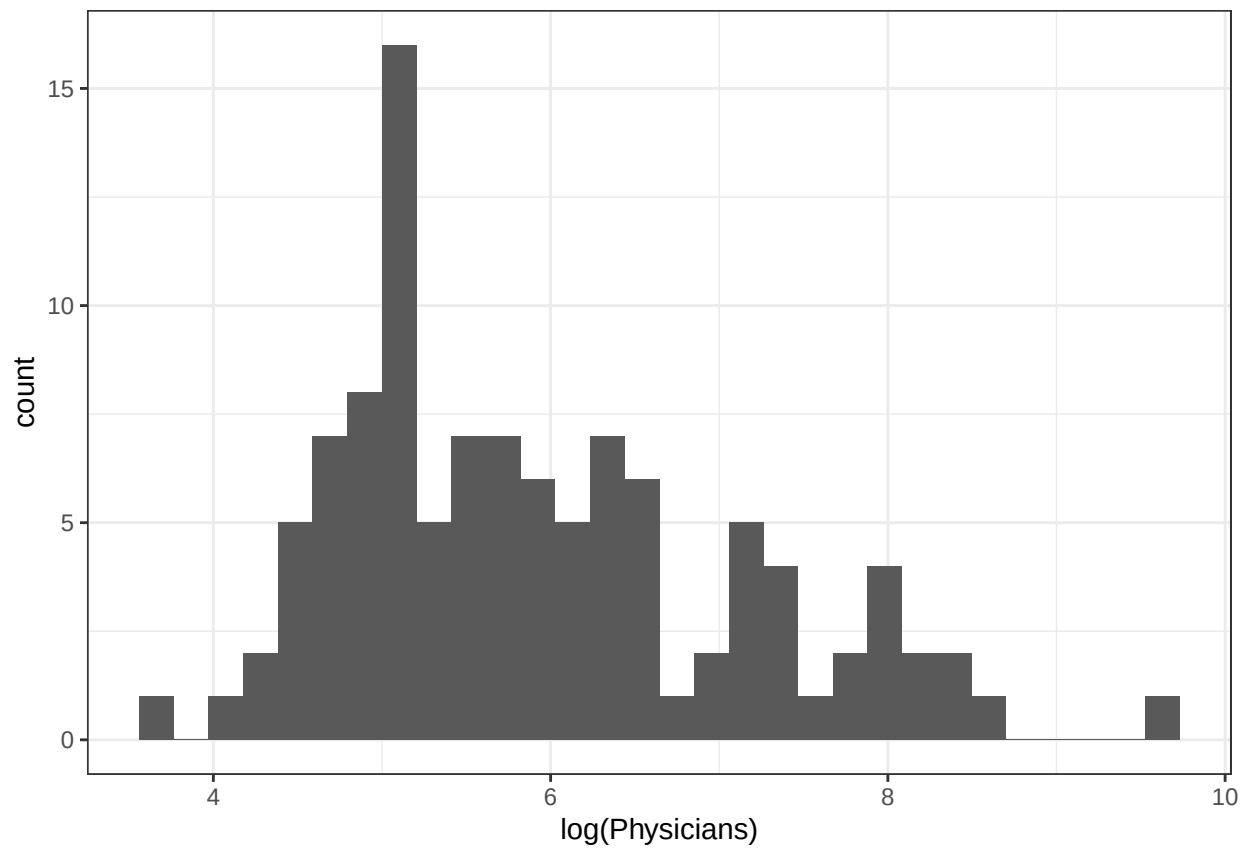
transformed model is larger than the original model ($p = 3.25e^{-05}$) but both models indicate the predictor variable being significant. The transformed CI [0.633, 2.053] is much larger than the original model CI [0.082, 0.21], this indicates that the transformed model may not be a better model than the original. When looking at both the Residuals vs. Fitted and Q-Q plots we can see that both the original and transformed models are very similar with slight variation but by looking at each plot we can see homoscedasticity and the residuals being normally distributed. So the transformed model did not explicitly improve the fit of the data compared to the original. Looking at the R^2 values, original ($R^2 = 0.1963$) and transformed ($R^2 = 0.0284$), we can see that the transformed model is worse at explaining the variation being caused by the predictor variable which is Population 18 to 34. Based off the values and plots, we will not choose the log transformed model to represent region 4 but stick with the original as the log transformation did not improve the model's ability to explain the relationship between total number of crimes and Population 18 to 34.

```
# Total Population
region1 |>
  ggplot(aes(x = log(total_population), y = LogY_i)) +
  geom_point() +
  theme_bw()
```

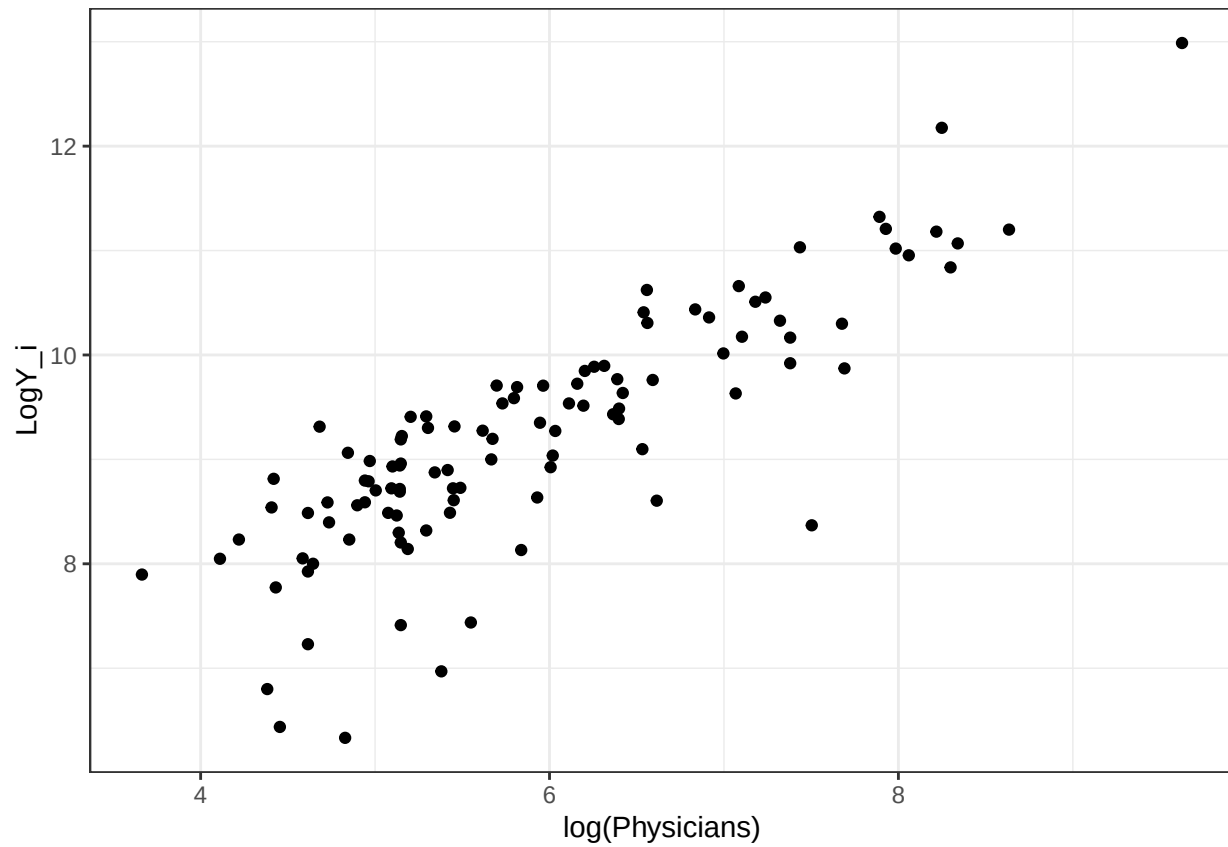


```
# Physicians
region2 |>
  ggplot(aes(x = log(Physicians))) +
  geom_histogram() +
  theme_bw()
```

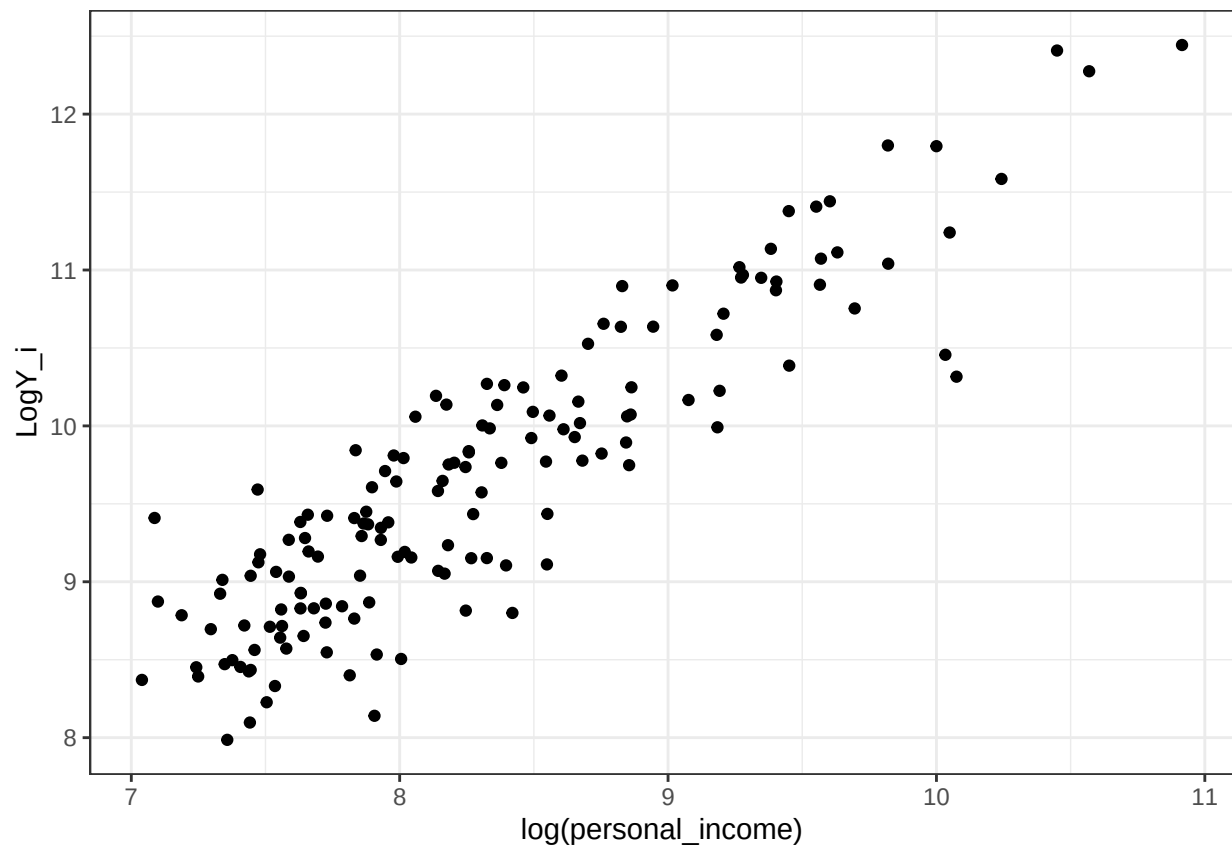
```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



```
region2 |>  
  ggplot(aes(x = log(Physicians), y = LogY_i)) +  
  geom_point() +  
  theme_bw()
```



```
# Personal Income  
region3 |>  
  ggplot(aes(x = log(personal_income), y = LogY_i)) +  
  geom_point() +  
  theme_bw()
```



```
# Population 18-34  
region4 |>  
  ggplot(aes(x = log(Population_18to34), y = LogY_i)) +  
  geom_point() +  
  theme_bw()
```

